Supplement 1

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Study protocol v4.0– earlier version

Global Health Research

Non-CTIMP Study Protocol

A mHealth intervention (mTB-Tobacco) for smoking cessation in people with tuberculosis: a two-stage adaptive design, randomised trial

Quit 4 TB Trial

Sponsor	The University of Edinburgh ACCORD The Queen's Medical Research Institute 47 Little France Crescent Edinburgh EH16 4TJ
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Funder	NIHR Global Health Research Unit on Respiratory Health (RESPIRE)
Funding Reference Number	NIHR132826
Chief Investigator	Professor Harry Campbell
Sponsor number	AC22139
Ethics Number	EMREC-RESPIRE-23-01
Trial registration number	ISRCTN86971818
Version Number and Date	V4.0 19th April 2023

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LIST OF ABBREVIATIONS

ACCORD	Academic and Clinical Central Office for Research & Development
CI	Chief Investigator
PI	Principal Investigator
CRF	Case Report Form
GCP	Good Clinical Practice
ICH	International Conference on Harmonisation
LMIC	Low and Middle-Income Country
DOTS	Direct Observation Treatment Strategy
QA	Quality Assurance
SOP	Standard Operating Procedure
TB	Tuberculosis

1 INTRODUCTION

1.1 BACKGROUND

Tuberculosis (TB) affects >10 million people every year and approx. 1.5 million die as a result.[1] While its incidence has declined in many parts of the world, the global TB pandemic continues to be fuelled by comorbid conditions (e.g. HIV, diabetes) and risk factors (e.g. smoking,[2] indoor air pollution[3]). Among these, smoking alone is responsible for 16% of the total TB disease-burden.[4] Smoking increases the chances of acquiring TB infection and TB disease by two- and three-folds, respectively;[5] continued smoking by those with active TB disease worsens their clinical outcomes.[3] Compared to non-smokers, TB patients who smoke are more likely to have severe forms of clinical presentation, poor bacteriological and clinical responses to treatment(s), and high rates of treatment failure.[6] Smoking doubles the risk of TB-related deaths[5] and, in survivors, TB recurrence.[7]

Smoking prevalence in TB patients is higher than the general population,[8] but the vast majority of TB patients are neither routinely asked about their smoking status nor advised to quit.[9] The strong association between smoking and TB has led to increasing recognition of the need for evidence-based smoking cessation approaches[10] and increasing policy support to help TB patients quit smoking.[11]

1.2 RATIONALE FOR STUDY

In a series of randomised-controlled trials (RCT)[10,12] in Bangladesh and Pakistan, our team has developed strong evidence to support face-to-face behavioural interventions achieving high quit rates.[10] We have also demonstrated that those who quit smoking show better clinical outcomes.[13] However, for several health system barriers[14] (cost, reach, sustainability) no TB high-burden country has so far integrated face-to-face smoking cessation within TB services. Recognising the challenges of integrating and scaling up face-to-face interventions, we have worked with WHO to develop a mHealth smoking cessation package (mTB-Tobacco) that can be delivered as SMS messages via mobile phones in TB patients.[15] While delivering mHealth package will cost less than face-to-face behavioural interventions, it is unknown if this package is as effective and therefore more cost-effective than face-to-face behavioural intervention. We have also consulted with TB programme managers in Bangladesh and Pakistan, colleagues working in TB DOTS Partnership and in International Union Against TB and Lung Diseases and evaluation of a mHealth package to achieve smoking cessation in TB patients has been identified as a priority.

2 STUDY OBJECTIVES

2.1 OBJECTIVES

2.1.1 Primary Objective

In people with TB who smoke daily, we will assess the effectiveness and cost effectiveness of mTB-Tobacco in achieving continuous abstinence for at least six months.

2.1.2 Secondary Objectives

In people with TB who smoke daily, we will assess the effectiveness and cost effectiveness of mTB-Tobacco in enhancing TB treatment adherence and improving clinical outcomes.

2.2 ENDPOINTS

2.2.1 Primary Endpoint

The primary endpoint will be biochemically verified continuous abstinence at 6 months post randomisation.[16] Abstinence is defined as self-report of not having used more than 5 cigarettes, bidis, water pipe sessions since the quit date, verified biochemically by a breath carbon monoxide (CO) reading of less than 10 ppm at month 6.

In case of concomitant smokeless tobacco use, biochemical verification will be carried out using Accutest® NicAlertTM strips to detect cotinine (a nicotine metabolite) level in urine samples. Reading of the strip will indicate a cotinine level (0-6), where level <3 (i.e., equivalent to 100-200ng/mL cotinine) will be considered tobacco abstinence.

When a patient self-reports abstinence with an elevated CO level of >10ppm (and cotinine levels in concomitant users indicating active tobacco use), the biochemical verification will supersede the self-report and the patient will be defined as a tobacco user.

2.2.2 Secondary Endpoints

Secondary outcomes will include:

1. Point abstinence, defined as a self-report of not using tobacco in the previous 7 days, assessed at week 9 and month 6.
2. Adherence to TB Treatment: All registered TB patients' medication logs (for anti-TB medication) are recorded on the 'Treatment Support Card'; a copy of this card would be requested from the TB paramedic by the RA and attached with the patient CRF.

3. TB Programme outcomes: the proportion of treatment success (including cured and completed treatment), treatment failure, defaulted and died will be recorded from TB register (TB03) at month 6. The definitions of these outcomes are as follows:
4. Cured: 'A patient who was initially smear-positive and who was smear negative in the last month of treatment (at month 6) and on at least one previous occasion'
5. Completed treatment: 'A patient who completed treatment (at month 6) but did not meet the criteria for cure or failure.'
6. Treatment failure: 'A patient wo was initially smear-positive and who remained smear-positive at month 5 or later during treatment'
7. Defaulted: 'A patient whose treatment was interrupted for 2 consecutive months or more'
8. Died: 'A patient who died from any cause during treatment'
9. Relapse: 'A patient who was previously treated for TB, was declared cured or treatment completed at the end of his most recent course of treatment, and is now diagnosed with a recurrent episode of TB (either a true relapse or a new episode of TB caused by reinfection)'

3 STUDY DESIGN

Using an adaptive design, we will conduct a multi-centre, cluster randomised, controlled trial consisting of four phases. Phase 1 involves consultation with Patients and Public Involvement (PPI) group about the study processes. PPI members will also review and provide feedback on the participant facing materials. Phase 2 is the pilot study. In Phase 3 **(superiority trial)**, which will last for 12 months (6 months recruitments and 6 months follow ups), we will compare mTB-Tobacco (intervention A) with usual care (control). In Phase 4 **(non-inferiority trial),** which will last for another 12 months (6 months recruitment and 6 months follow ups), we will compare mTB-Tobacco (intervention A) with face-to-face behavioural support (intervention B). See QUIT 4 TB Trial flow diagram 1 below for details.

4 STUDY POPULATION

4.1 STUDY LOCATION

We will conduct the trial in two countries with high TB and tobacco-disease burden:

	TB (per 100,000)	TB cases (n)	TB deaths (n)	Tobacco prevalence
Bangladesh	221	357,000	47,000	35%
Pakistan	265	562,000	44,000	20%

We will recruit TB health facilities (clusters) in the public sector from Dhaka division in Bangladesh and Punjab province in Pakistan.

4.2 SAMPLE SIZE

The study requires a total of 2,384 smokers with TB (~50 recruits from **44 health facilities** [the clusters]) assuming that 10% do not provide primary outcome data. This sample also includes the first 16 participants taking part in the pilot study in case the intervention needed major refinement and we are unable to use their data. Phase 3 comparing A with C, would have 90% power at 5% significance level to compare 16 clusters randomised to A with 8 clusters randomised to C, assuming proportion achieving abstinence in usual care was 8% and using the mTB-Tobacco was 18% abstinence at 6 months. If A was superior to C, Phase 4 would then continue, and would have 90% power at a 1-sided 2.5% level to establish non-inferiority of mobile to face-to-face, comparing 20 clusters in each of A and B assuming face-to-face achieved 18% abstinence at six months and the non-inferiority margin was 8%. Assuming that just 2% would give up with no intervention at all, the established treatment effect of face-to-face over natural cessation over 6 months would then be 16% (18%-2%). The non-inferiority margin of 8% equates to the mobile intervention preserving at least 50% (8/16) of the established treatment effect of the face-to-face intervention. We will check above sample size assumptions mid-way through the recruitment period and adjust the sample size accordingly.

4.3 INCLUSION CRITERIA

4.3.1 Clusters (health facilities) inclusion criteria

All TB health facilities with a minimum turnover of 50 new TB patients a month will be eligible. The included sites must be functioning, and designated TB diagnostic centres approved by the National /Provincial TB control Programme (NTP/PTP). These may include primary, secondary, or tertiary care health centres (HCs), generally known as the district/or tehsil headquarter hospitals in Pakistan. In Bangladesh, health facilities will be the DOTS corners at Upazila (sub-district) Health Complex and Urban DOTS corners. In addition, the following conditions must be met:

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- The facility must be able to nominate a site focal person (an NTP personnel e.g., the district TB coordinator, or the responsible TB clinician at the site) who assumes responsibility for the proper conduct of the trial at their site.
- The HC should have available an adequate number of qualified staff and adequate facilities for the foreseen duration of the trial to conduct the trial properly and safely.
- The HC should be willing to join the trial and provide required administrative support.

4.3.2 Participants inclusion criteria

Patients will be considered eligible for enrolment in this trial if they fulfil all the following inclusion criteria and none of the exclusion criteria (see 4.4.2).

- Age at least 15 years (counted as adult TB patients in Bangladesh and Pakistan)
- Willing and able to provide written informed consent
- Diagnosed with drug sensitive PTB (smear positive or negative) in the last four weeks
- Currently smokes tobacco on a daily basis or have only stopped or reduced smoking (less than daily) since diagnosed with TB
- Willing to quit tobacco use
- Access to personal mobile phone

4.4 EXCLUSION CRITERIA

4.4.1 Clusters (health facilities) exclusion criteria

Facilities not meeting the inclusion criteria will not be eligible for the trial.

4.4.2 Participants exclusion criteria

Patients anticipated to have adverse effects either due to the study treatment or research burden, will be excluded. These will include those who are/have:

- Less than 15 years of age
- Retreatment TB, MDR TB, Miliary or Extrapulmonary TB
- Currently using any pharmacotherapy for tobacco dependence
- Unwilling or unable to provide written informed consent

5 PARTICIPANT SELECTION AND ENROLMENT

5.1 IDENTIFYING PARTICIPANTS

Doctors, paramedic/nurses, Health Workers (HW), other clinical staff and laboratory technicians engaged in TB care and management at the HC will identify TB patients. It is anticipated that the bulk of the patients will be identified through the TB registers kept by the TB paramedics (aka TB DOTS facilitators), responsible for recording, reporting, and dispensing anti-TB medications to the patients. Once a patient is identified, the resident Research Assistant (RA) will complete the Screening Form for assessing eligibility.

5.2 CONSENTING PARTICIPANTS

Eligible patients will be given verbal and written information about the trial and 24 hours to consider participation. They will have the opportunity to discuss the trial fully with the RA (and the HW/TB paramedic if patient requests). Each participant will be informed of the aims, methods, anticipated benefits, potential hazards, and discomforts of the study, both through the Patient Information Sheet and verbally. We will be using the same procedures of obtaining consent as in the previous trial[17] sponsored by the University of York. The research assistant (RA) will be trained to obtain informed consent from the participants. There is unlikely to be information loss unless it is done in a rush. The main points will be summarised to bullet points for the RA to read to the participants, which will reduce the scope of information loss to a minimum. The RA will also check orally with the participants if they have understood the key information. The patient's right to decline participation and the right to withdraw at any time, without the need to give an explanation and without it being detrimental to their overall treatment will be clearly stated. Each participant will be reimbursed for their travel expenses (about 200PKR/visit; 113BDT/visit; £0.97/visit). Written consent will be obtained from those interested by the RA – trained (by the local PIs and Research Fellows/Trial Coordinators) on the appropriate information and consent procedures for the TB patients (prior to the start of patient enrolment in study). All participants agreeing to participate will be required to sign (or place a thumbprint using indelible ink) the consent form alongside their National Identity Card number (if any), address and mobile telephone number.

5.2.1 All consent forms will be stored in accordance with local requirements. A copy of the consent form will be given to the patient and the original signed copy kept in a locked cabinet in the respective research offices. The Research Fellows/Trial Coordinators will bring the original copy to the corresponding country coordinating centre for central monitoring purposes. Withdrawal of Study Participants

Participants are free to withdraw from the study at any point or a participant can be withdrawn by the Principal Investigator. The Principal Investigator (PI) may withdraw a participant if it is considered in the participant's best interest not to continue the study e.g. if participant's health has deteriorated and their continued participation may undermined their health further. In rare situation that the PI decides to withdraw a participant, they will check with other clinicians. If withdrawal occurs, the primary reason for withdrawal will be documented in the participant's case report form, if possible. If a participant decides to withdraw, they will be informed that:

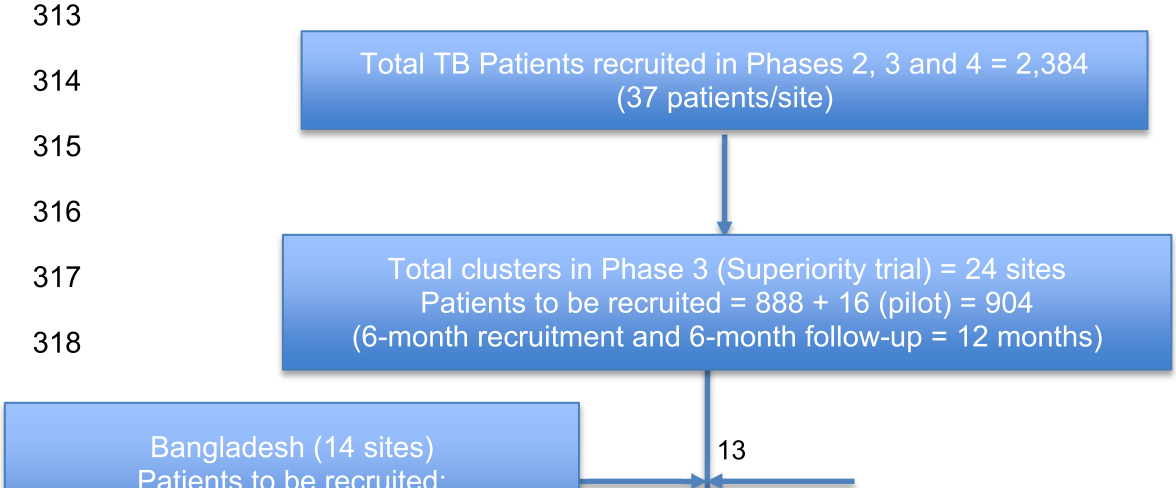
a. They will be withdrawn from all aspects of the trial but continued use of data (and samples) collected up to that point.

To safeguard rights, the minimum personally identifiable information possible will be collected.

Detail reasons and procedures for a study participant stopping early i.e., "stopping rules" and "discontinuation criteria"

6 RANDOMISATION AND ALLOCATION PROCEDURES

Since the interventions will be delivered at a group level, this trial is a cRCT where health facilities will be randomly allocated to the different trial arms. The health facilities (clusters) would be randomised 2:2:1 to mTB-Tobacco (intervention A), face-to-face (intervention B), or standard care (intervention C). An independent statistician based at University of Edinburgh who will be blinded to centres, will identify and use computer-generated random-number lists to generate the allocation sequence. The allocation will be based on using minimisation to ensure balance across the groups on the average number of TB patients seen per month and geographical location (Bangladesh and Pakistan).



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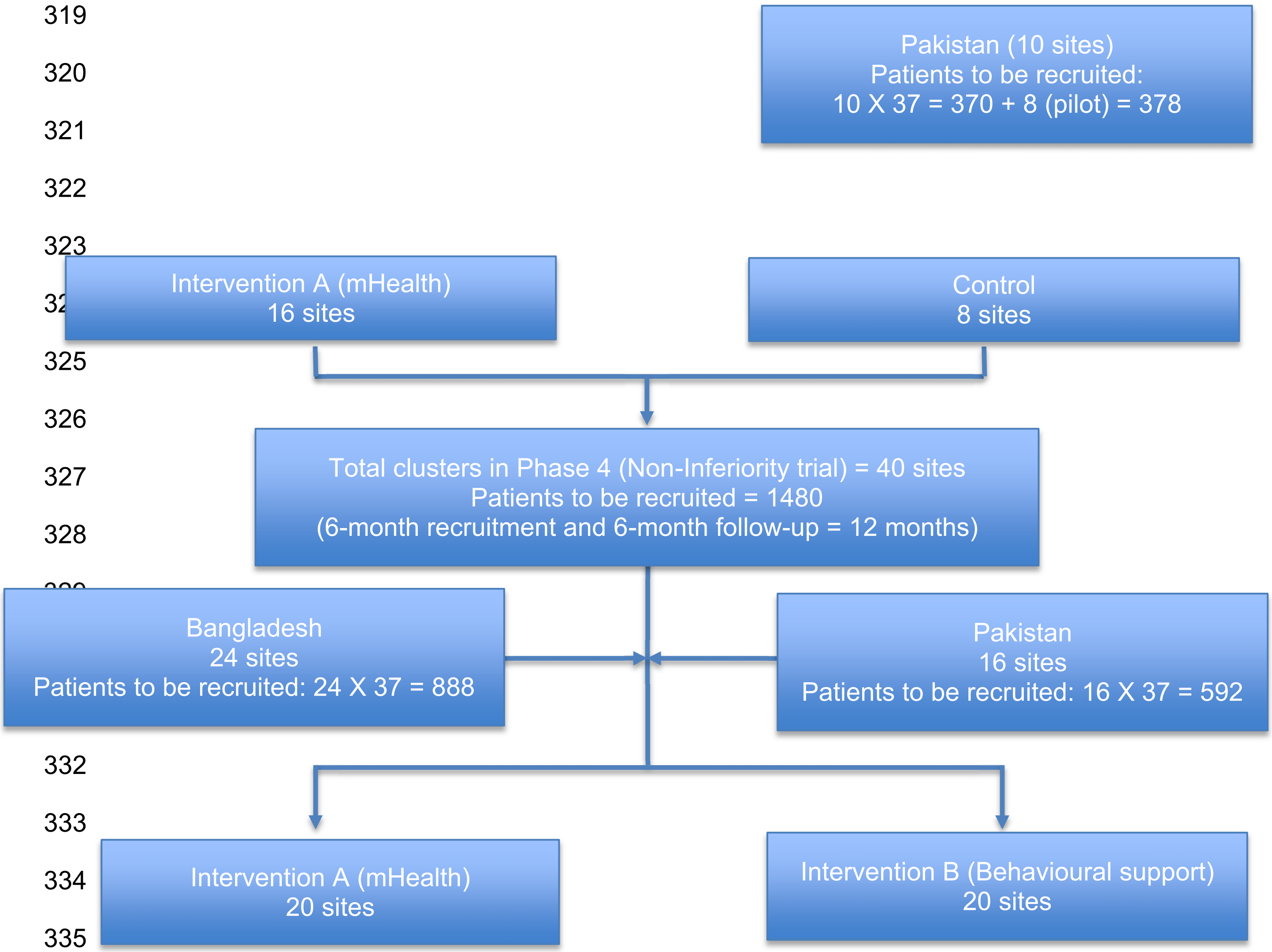

Diagram 1: Quit 4 TB Trial flow diagram

7 INTERVENTIONS

7.1 mTB-TOBACCO (INTERVENTION A)

The mTB-Tobacco programme (intervention A) is designed to send short message service (SMS) content with two distinct areas of intervention. The first tries to instil behaviour change in patients to quit the habit of tobacco use and the second includes a combination of supportive motivational and informative messages through the period of TB treatment. The overall outcome being that a person is able to both successfully quit tobacco consumption and complete TB treatment.

A generic intervention has already been developed by the investigators in collaboration with the WHO mHealth team, available here: https://www.who.int/publications/i/item/a-handbook-on-how-to-implement-mtb-tobacco. This resource provides guidance for national TB control programmes and organisations responsible for delivering TB control in adapting and implementing a bespoke mTB-Tobacco programme. The resource includes a set of SMS messages to inform TB patients about the hazards of tobacco use and encourage them to quit. Furthermore, these messages are also designed to enhance treatment adherence, increase healthy behaviours, and reduce potentially harmful behaviours in TB patients in order to improve their health outcomes.

To make the intervention ready for the trial, the research team will set up a reference group (TB and tobacco experts, TB programme managers and clinicians) to support, and inform the adaptation of mTB-Tobacco intervention. The group will first select a set of SMS messages (from the abovementioned resource) and draft their transmission schedule (the number of messages, their frequency and time intervals). Once a plan of the mTB-Tobacco programme has been developed, it will go through iterative stages of pre-testing, piloting, and refinement.

In phase 1, we will organise PPI forum meetings to ask them to review our patient facing materials including SMS messages for comprehension, language, and tone. The PPI group will comprise of six-eight members including TB patients, carers and lay members of public, both in Bangladesh and Pakistan. Messages that are considered as unclear or inappropriate for the target group would be rewritten.

Once the messages have been reviewed by the PPI group, the intervention will be tested (as part of the pilot study) with a total of 16 patients (8 each in Bangladesh and Pakistan). The pilot will be evaluated using two sets of data: users' self-reported experiences and their real-time engagement with the programme. Users' experiences can be collected through face-to-face/phone-based interactions. Key questions would focus on users' experience of taking part in the mTB-Tobacco programme; the clarity, quantity, timing and frequency of messages; what was good about the programme and what was not; completion or non-completion the programme; and any effect on their attitudes or target behaviour(s). Users' real-time engagement will be evaluated using computer records from the programme used for launching mTB-Tobacco. The pilot study is embedded in the superiority trial (phase 3). The pilot participant data will be treated with the same processes and governance as the main trial participant data.

Once mTB-Tobacco is piloted, but before continuing to superiority trial, the programme will go through a refinement process (a study pause) that will address issues highlighted during the pilot and make adjustments to the intervention, if needed.

At the end of above exercise, the intervention will be ready for the superiority trial and will comprise of: (a) a set of SMS messages translated in Bangla and Urdu, respectively; (b) a schedule of transmission; (c) an automated digital SMS messaging platform ready to send these messages as per schedule; and (d) a dedicated team to supervise and monitor the transmission of these messages. The SMS messaging will only store unique ID number for the trial. No other personal or identifiable information will be stored. Each participant in the intervention group will receive a total of 178 SMS messages over a period of 6 months. In the first two months, the frequency of messages would be 4 to 5 messages per day, in next two month the frequency would reduce to 2 to 3 messages per day and in the last two months there will be 1 to 2 messages sent per week.

We are using REDCap for managing the data for this trial. The mHealth app will be linked to REDCap. At the screening stage, participants who meet all inclusion criteria and do not meet any exclusion criteria will be eligible to take part in the study. They will be given the participant information sheet and consent form. If they consent to take part in the study, their demographic details (including active mobile number) will be recorded on the REDCap database. If the participant is marked as belonging to the intervention arm, then his/her contact details/mobile number will be captured from the database by the mobile SMS service to send mHealth messages as per the schedule. The participants belonging to the intervention arm will start receiving the SMS messages, whereas those belonging to control arm will not receive any SMS.

The mobile SMS service is provided by a third-party service. A contract of services for the development of the mTB-Tobacco program application has been signed between The Initiative and Biter Tech (a third-party). Private Limited. Section A General Terms & Conditions Point 7 of the sub-contract (highlighted in yellow) mentions that “The Consultant is not allowed to keep or sell any project related data. The Consultant agrees to keep confidentiality of project data, and, to cause its employees and agents to keep confidentiality of any data available to them. The consultant will not access or check participant’s personal data unless it is necessary in connection with the services provided under this Agreement, or otherwise consented to, in writing.”

7.2 FACE-2-FACE BEHAVIOURAL SUPPORT (INTERVENTION B)

This consists of an adapted version of a pre-developed and proven effective intervention to help people to quit smoking and smokeless tobacco use.[10] This consists of two face-to-face sessions delivered at day 0 and day 5 (+2) and which last 10 and 5 minutes respectively. The sessions will be structured using an educational flipbook; the session on day 0 will be aimed at encouraging tobacco users to see themselves as non-users and set a plan for a quit date five days later; with a session on the quit date (day 5) to review progress. Further encouragement and support (if needed) will be offered at a subsequent visit at week 5.

7.3 STANDARDISED USUAL CARE (INTERVENTION C)

The TB health professionals will be provided with patient education leaflets to be given out to all patients allocated to the respective trial arm. Leaflets for patients and others in the house contain information on the harmful effects of tobacco and advice on stopping its use.

8 STUDY ASSESSMENTS

8.1 STUDY ASSESSMENTS

All patients will have their follow-ups flagged up (in the trial database that can be printed off) to avoid attrition, both at the recruiting site and the respective country coordinating centre. The patients will have assessments on day 0, week 9 and month 6 time-points correspond with routine TB clinic visits. The assessment schedule is tabled below.

Assessment	Screening	Day 0 baseline	Week 9	Month 6
Assessment of Eligibility Criteria	☒			
Written informed consent	☒			
Face to face / phone-based interaction*			☒	
Self-reported tobacco use/abstinence status		☒	☒	☒
CO measurement				☒
Socio-Demographic information		☒		
Tobacco use and quit history		☒		
Nicotine dependency		☒	☒	☒
Economic outcomes		☒		☒
Process outcomes (Dose delivered and fidelity)			☒	☒
Medication (TB) compliance			☒	☒
Adverse events review			☒	☒
TB outcomes				☒

* This applies to Phase 2 pilot participants only.

The trial assessment schedule will outline visit dates (with windows) necessary for data collection, but the patient may be seen more frequently for clinical care as needed.

During the treatment period, if patients are unable to attend on the day, every effort should be made to complete the visit within 2 days of the scheduled date. If a scheduled visit is missed without notice, then the RA should endeavour to contact the patient by phone or by home visit.

During the follow-up phase, scheduled assessments should be carried out no more than 5 days before the scheduled visit. HCs may choose to re-schedule visits to allow for public holidays or other unavoidable circumstances that affect the scheduled visit date, but the re-scheduled visit should be no more than 5 days from the originally scheduled visit date. Patients will also be given a card with the contact details for the trial RA and the clinical TB care team at their site.

If a patient is more than 5 days late for a scheduled study visit, an additional visit will be performed as soon as possible, including the appropriate assessments that were specified in the trial schedule for the visit that was missed.

9 DATA COLLECTION

This is a cRCT collecting quantitative data only. Following types of data will be collected:

- Personal identifiable data – this will include name, address, mobile telephone number and the national identity card number (if any) and will be stored together with the consent forms separate from other sources of data.
- Socio-demographic data – this will be collected only once at baseline using the Case Report Forms.
- Self-reported data on tobacco use behaviour (past and present) including quitting or staying abstinent, nicotine dependence and withdrawal symptoms – this will be collected only once at baseline using the Case Report Forms.
- We will also collect economic data which will include cost of the delivery of intervention collected from various sources (see economic analysis subsection), health service use and quality of life data collected through Case Report Forms.
- Process data will be collected to assess the fidelity to the delivery and receipt of SMS and face-2-face messages as part of mTB-Tobacco and behavioural support interventions, respectively. mTB-Tobacco fidelity data will be collected from the digital platform delivering messages and behavioural support fidelity data will be collected using the Case Report Form.

- Biochemical verification using CO monitors will be conducted for all those who report abstinence from tobacco use. This will be collected at the final follow using the Case Report Forms.
- TB medication adherence and TB outcomes data will be collected from routine TB data collected and recorded in TB registers, but this will also be validated through the Case Report Forms.

The data collection will consist of responses to a set of structured questions in the Case Report Forms which will be completed and facilitated by the field team. Tablets will be used to enter data directly, which will appear in the database in real time. The data entered will synchronise with the database at the University of Edinburgh on a secured server. All the de-identified data will be transferred to the statistician in Edinburgh who will conduct the analysis using the statistical software packages: STATA and R.

Overall data quality will be ensured through training and supervision. Data entry validation will occur by double-checking a random sample of the data in the field. Moreover, quantitative data once entered will also be peer-reviewed by the Edinburgh-based statistician. We have used these processes successfully in previous studies in Pakistan and Bangladesh.

9.1 SOURCE DATA DOCUMENTATION

The following data should be verifiable from source documents:

- All signed consent forms
- Dates of visits
- Eligibility and baseline values for all participants
- All clinical endpoints (including TB related to the trial)
- Routine patient clinical and laboratory data
- Medication compliance
- Collection of personal data
- Written notes for face to face / phone-based interactions with Phase 2 pilot participants at week 9 (will be used to inform the protocol for Phases 3 and 4 of the study)

All information collected during the course of the study will be kept strictly confidential. The researchers will also comply with all aspects of the UK General Data Protection Regulation (GDPR), the Data Protection Act 2018, and the Privacy and Electronic Communications (EC Directive) Regulations 2003, and operationally this will include:

Consent from patients to record personal details including name, date of birth, and address.

Appropriate storage, restricted access, and disposal arrangements for patients' personal details.

Identifiable information will be collected on the consent form when he or she is consented into the study, but all other data transferred to or from the country coordinating centre will be coded with a Trial number and will not include patients' identifiers.

If a patient withdraws consent from further study participation, no further assessments will be conducted but their data already collected will remain on file and will be included in the final study analysis.

Data will be securely archived by University of Edinburgh for a minimum of 10 years. Following authorisation from the sponsor, arrangements for confidential destruction will then be made.

9.2 CASE REPORT FORMS

A validated case report form will be used to collect all participants' data at the baseline and follow ups. This will be captured digitally using an app on a password-protected tablet computer.

10 DATA MANAGEMENT

A trial specific data management plan (included in the Manual of Procedures (MoP)) agreed by the Chief Investigator, trial unit and other study investigators will be drafted to provide detailed instructions and guidance relevant to database set up, data entry, validation, review, query generation and resolution, quality control processes involving data access and transfer of data to the University of Edinburgh at the end of the study and archiving.

Participants' name and personal identifiers will be removed. All information will be coded and de-identified. The information we have collected in paper copies will be stored under lock and key cabinet at

The Initiative,
Main Office
Orange Grove Farm, Main Korung Rd
Banigala, Islamabad
Pakistan

AKR Foundation
Suite C-3 & C-4,
House #06,
Road # 109,
Gulshan-2, Dhaka-1212,
Bangladesh

The electronic data will be stored in a secured computer which can only be accessed with a secure password. Only the researchers will have access to the data.

10.1 DIRECT ACCESS TO SOURCE DATA

A statement of permission to access source data by study staff and for regulatory and audit purposes will be included within the patient consent form with explicit explanation as part of the consent process and Participant Information Sheet. Once pseudonymised, data will only be shared between respective country coordinating centres and the University of Edinburgh. Once the team has completed the analysis and published all intended scientific journals, it will be made available for other researchers.

In principle, pseudonymised data will be made available for meta-analysis and where requested by other authorised researchers and journals for publication purposes. Requests by other researchers for access to pseudonymised data archived on Edinburgh DataVault will be reviewed by the Principal Investigators. Edinburgh DataShare is an open access research data repository.

The Investigator(s)/Institutions will permit monitoring, and REC review (as applicable) and provide direct access to source data and documents.

Study data will be recorded in a number of files for both the administration of the study and collection of patient data.

All data will be completely pseudonymised for purposes of analysis and any subsequent reports or publications. For the purposes of ongoing data management, once randomised, individual patients will only be identified by Trial numbers, CRFs will be designed using ReDCAP software. The CRFs will be available as an online eCRF using a web-based interface.

10.1.1 Data entry

The data collected will be entered into a secure web-based interface, specifically developed for this study. Confirmation of the data received by the trial's unit will be given to the country trial manager.

All data will be stored and transferred following Health Insurance Portability and Accountability Act (HIPAA). HIPAA standards either comply with or exceed data protection regulations in operation in the two study countries, Bangladesh and Pakistan. Therefore, adherence to HIPAA ensures that sufficient data management standards are met for this international study. The staff involved in the trial (both at the sites and country coordinating centres) will receive training on data protection. The staff will be monitored to ensure compliance with privacy standards. Data will be checked according to procedures detailed in the trial specific Data Management Plan.

10.1.2 Data storage

Each site will hold data according to the Data Protection Act 2018 and data will be collated in CRFs identified by a unique identification number (i.e. the Trial number) only. For example, the Trial Screening Forms will contain a number to identify the site and a screening number (which will be the patient TB registration number- see section 3.3.1 on screening procedures for details) unique to each patient at that site, and once randomised, follow up forms will contain a Trial number unique to each patient in the trial. An Intervention Allocation Log at the coordinating centres will list the TB registration number of the patient against allocated Trial number for those randomised. The trial database will maintain a list of all TB registration numbers and Trial numbers for all trial patients at each site for the entire trial.

All study files will be stored in accordance with GCP guidelines. Electronic data will be password protected and stored on a secure server by the trial's unit. Only participants' identification numbers (site number, TB Registration number, Trial number) will be used to uniquely identify patients on the online electronic eCRFs.

Study documents (paper and electronic) at the participating sites, the country coordinating centres and the University of Edinburgh will be retained in a secure (locked when not in use) location, during and after the trial has finished. All personal data including identifying information will be stored in a secure data-storing facility and secure servers for a minimum of 10 years following the end of the study at:

The Initiative,
Main Office

Orange Grove Farm, Main Korung Rd
Banigala, Islamabad
Pakistan

AKR Foundation
Suite C-3 & C-4,
House #06,
Road # 109,
Gulshan-2, Dhaka-1212,
Bangladesh

Data will be transferred from Bangladesh and Pakistan to University of Edinburgh in the UK via a secure, password-protected online platform (DataSync) and then stored on a secure, password-protected DataVault / DataShare at University of Edinburgh for a minimum of 10 years following the end of the study. After 10 years, all paper records will be shredded and disposed of securely and electronic records will be permanently erased. All essential documents, including source documents, will be retained for a minimum period of 10 years after study completion.

The University of Edinburgh has a backup procedure approved by auditors for disaster recovery. There will be a separate archival of electronic data performed at the end of the trial, to safeguard the data, and in accordance with regulatory requirements.

10.2 QUALITY ASSURANCE AND CONTROL

10.2.1 Training

The following training procedures will be conducted to ensure quality control.

Person trained	Description
RA, Clinical TB care team	• CO/Cotinine testing • Consent procedure • Data collection and management • Safety and effectiveness assessments • Eligibility assessment
Country PI/ TM	Randomisation, Database management (data entry and querying procedures), management of logs/files/registers kept at the coordinating centre

10.2.2 CO and Cotinine testing

Data quality for CO and cotinine testing will be ensured through training, supervision and calibrating CO monitors. In this trial, the RA will perform cotinine dipstick tests on urine samples collected from the patients, which will be discarded after cotinine levels have been recorded on the CRFs.

10.2.3 Site Approval, Start-up Procedures and Ongoing Support

The following documentation must be in place and verified by the country PI's in order for a HC to be eligible to participate in the trial and to begin recruiting:

- A copy of signed Approval letter or Memorandum Of Understanding (MOU)
- A copy of the IRB approval (in line with local regulatory requirements)
- Completed Delegation of responsibilities Log and Contact Details (this could be incorporated in the MOU as per local procedures)

Once this documentation has been received, confirmation that the study may commence at that site will be sent to the site PI by the country PI.

Prior to commencing the study, each study site will be visited by the country TM or PI.

RA and the Clinical TB care team at the site will be trained in the data collection, data transfer, and filing and other trial procedures in order to comply with GCP. At this time the protocol will be reviewed in detail and the MOP provided.

Phone support, and visit support, will be provided to sites as per their requirements during the course of the study, mainly by the country TMs, with the support of the trials unit as appropriate.

10.3 DATA MONITORING

10.3.1 Central Data Monitoring at Edinburgh

Each country coordinating centre will be responsible for its own data entry and local trial management. Data will be entered into the trial database at the country coordinating centres. The sites/countries will retain the original CRFs. Data stored on the central database will be checked by personnel authorised by the trials unit for missing or unusual values (range checks) and checked for consistency within participants over time. If any such problems are identified, the country coordinating centres will be contacted and asked to verify or correct the entry. Changes will be made on the original CRF at the site/country and entered into the database at the country coordinating centre.

10.3.2 In-country Data Monitoring

The country coordinating centres will send reminders for any overdue and/or missing data with the regular inconsistency reports of errors to the RA at each participating site.

The country trial manager and/or PI will visit the trial sites to validate and monitor data. The frequency, type and intensity for routine monitoring and the requirements for triggered monitoring will be detailed in the trial specific operating procedures. The Quality Assurance plan will also detail the procedures for review and sign-off.

A detailed site initiation visit with training will be performed at each study site by the country PI or designated trial manager. The site initiation visits will include training in the study protocol, consent processes, and administration of the study interventions.

Monitoring and site training will be carried out at each site within intervals specified in the 'Quality Assurance Plan'. The monitoring will adhere to the principles of ICH GCP and will follow an agreed monitoring plan.

Delegated members of the research team will:

- confirm adherence to protocol
- review eligibility verification and consent procedures
- look for missed outcomes reporting
- verify completeness, consistency and accuracy of data being entered on CRFs
- evaluate medication accountability
- provide additional training as needed

The delegated members of the research team will require access to all (enrolled TB) patient medical records including, but not limited to, laboratory test results and prescriptions. The site PI (or delegated personnel) should work with the delegated members of the research team to ensure that any problems detected are resolved.

10.4 DIRECT ACCESS TO PATIENT RECORDS

Site PIs should agree to allow trial-related monitoring by providing direct access to source data and documents as required. Such information will be treated as strictly confidential and will in no circumstances be made publicly available data controller along with any other entities involved in delivering the study that may be a data controller in accordance with applicable laws (e.g. the research site or partnering institution in the local country where the study is being conducted)

10.4.1 Data Breaches

Any data breaches will be reported to the University of Edinburgh Data Protection Officer who will onward report to the relevant authority according to the appropriate timelines if required. These Data Protection Officers in Bangladesh and Pakistan will also be notified of data breaches:

Bangladesh
Ms Samina Huque
Database Manager
ARK Foundation
Bangladesh
Email: huque1983@gmail.com

Pakistan
Mr Saeed Ansari
Database Manager
The Initiative
Pakistan
Saaed.ansaari@gmail.com

11 STATISTICS AND DATA ANALYSIS

11.1 SAMPLE SIZE CALCULATION

The study requires a total of 2,384 smokers with TB (~50 recruits from 48 health clinics [the clusters]) assuming that 10% do not provide primary outcome data. This includes the first 16 participants for the pilot study. Phase 3 comparing A with C, would have 90% power at 5% significance level to compare 16 clusters randomised to A with 8 clusters randomised to C, assuming proportion achieving abstinence in usual care was 8% and using the mTB-Tobacco was 18% abstinence at 6 months. If A was superior to C, Phase 4 would then continue, and would have 90% power at a 1-sided 2.5% level to establish non-inferiority of mobile to face-to-face, comparing 20 clusters in each of A and B assuming face-to-face achieved 18% abstinence at six months and the non-inferiority margin was 8%. Assuming that just 2% would give up with no intervention at all, the established treatment effect of face-to-face over natural cessation over 6 months would then be 16% (18%-2%). The non-inferiority margin of 8% equates to the mobile intervention preserving at least 50% (8/16) of the established treatment effect of face-to-face. We will check above sample size assumptions mid-way through the recruitment period and adjust the sample size accordingly.

11.2 PROPOSED ANALYSES

11.2.1 Cost-effectiveness analysis

A health economic evaluation is included alongside the two phases to establish the value for money afforded by the mTB-Tobacco intervention. The cost of providing the interventions is recorded and calculated using local cost data. These costs include staff, premises and pharmaceutical products. We also record the wider use of public and private sector health care used by patients with a self-report service use questionnaire. A cost profile is calculated by applying country specific unit cost tariffs to quantities. Total intervention cost, costs of wider health care use and also patients' out of pocket expenditure are derived and a total cost applied to each patient in each trial arm.

In Phase 3 we will conduct an incremental cost-effectiveness analysis of mTB-Tobacco over and above usual care. Health Related Quality of Life (HRQoL) data will be collected using EQ-5D-5L to calculate Quality Adjusted Life Years (QALYs). Incremental cost effectiveness ratios will be presented in terms of i) cost per QALY and ii) cost per additional quitter as based on the secondary outcome of the trial. A 12-month time horizon is selected for the CEA in order to utilise the longer time window of health care use and changes in HRQoL. A full sensitivity analysis will be presented to assess the robustness of the results and the probability that mTB-Tobacco is cost-effective at a range of threshold values.

In Phase 4 (non-inferiority trial) we will record resource quantities and attribute unit costs to the mTB-Tobacco intervention and face-to-face behaviour support, inclusive of wider health system costs following the methodology of Phase 3. Costs for each arm are calculated and compared to a pre-defined acceptable cost difference. We will also carry out a cost-effectiveness analysis in Phase 4. Difference in costs and QALYs between treatment arms will be estimated to calculate the ICER based on cost per QALY and cost per additional quitter.

Underlying uncertainty around the decision to adopt the intervention will be assessed using non-parametric bootstrap re-sampling technique. Bootstrapping is an efficient method for calculating the confidence limits for the ICER as its validity does not depend on any specific form of underlying distribution. We will perform the bootstrap 5,000 replications and construct the 95% confidence intervals for the ICERs based on the bootstrapping results. Cost-effectiveness acceptability curves (CEACs) will be constructed based on the bootstrap iterations to estimate the probability that the intervention is cost-effective at different threshold values for one QALY.

The ICER, calculated in terms of the cost per QALY, will be used to assess the value of money afforded by the intervention over and above the control and draw conclusions with respect to the potential cost-effectiveness of the intervention.

To assess the impact of imputing missing data, we will also conduct a complete case analysis based on the participants who have both complete costs and QALYs at all timepoints, following the same analysis method as the primary analysis above. We will also conduct sensitivity analyses using pattern mixture modelling to examine the assumptions for multiple imputation methods.

11.2.2 Statistical analysis

Outcomes will be primarily analysed in a missing-at-random architecture by fitting generalised linear mixed models using a log link and assuming an underlying Poisson distribution with clusters treated as a random effect. All analyses will account for clustering through the random intercept model. The crude and adjusted relative risks (RRs) and corresponding 95% CIs will be estimated.

12. SAMPLE HANDLING, PROCESSING AND ANALYSIS

Urine sample will be collected from patients on their follow up visit who are using beside smoking other forms of tobacco (smokeless etc). The participants will be provided with containers and private premises to provide the sample. Nicotine dipstick test will be carried out by the research assistants on the sample and result recorded. The sample will then be discarded, hence no storage or shipped from research site to 3rd party/other institution/a UK based institution/ company is required. The sample analysis is not critical to the conduct of the trial and neither related to tertiary/exploratory endpoint data and objectives.

13. TRANSLATION

13.1 Translation Services

All trial documents (questionnaires, consent forms, participant information sheets, flipbook, mhealth messages) will be translated in-house. The participant-facing documents and SMS messages will be translated into local languages (Bengali for Bangladesh site and Urdu for Pakistan site) by the country-specific central research team. Then, another two research team members (one for each study site) will translate the local language versions back into English. Another two research team members (one for teach study site) will compare the back-translated versions with the original version for any discrepancies, and will resolve them before finalising the translated local versions. We will check the local language versions with the PPI group before they are submitted to the local ethics committees for approval.

14. ADVERSE EVENTS

In the context of this study, we think it is very unlikely that any Serious Adverse Event (SAE) would be related to the intervention or research procedures. The interventions are behavioural interventions and are not harmful. CO measurement and the quantitative data collection methods proposed present very minimal risks, if any risk, to participants. Therefore, no adverse events will be reported to the sponsor for this study

12 OVERSIGHT ARRANGEMENTS

12.1 INSPECTION OF RECORDS

Investigators and institutions involved in the study will permit trial related monitoring on behalf of the sponsor, ethics committees review, and regulatory inspection(s). In the event of monitoring, the Investigator agrees to allow the representatives of the sponsor direct access to all study records and source documentation. In the event of regulatory inspection, the Investigator agrees to allow inspectors direct access to all study records and source documentation.

12.2 STUDY MONITORING AND AUDIT

The Sponsor will assess the study to determine if an independent risk assessment is required. If required, the independent risk assessment will be carried out by the ACCORD Quality Assurance Group to determine if an audit should be performed before/during/after the study and, if so, at what frequency.

Risk assessment, if required, will determine if audit by the ACCORD QA group is required. Should audit be required, details will be captured in an audit plan. Audit of Principal

Investigator research sites, study management activities and study collaborative units, facilities and 3rd parties may be performed.

12.3 STEERING COMMITTEE/DATA MONITORING COMMITTEE

Due to the low-risk nature of this trial an independent steering and monitoring committee will be set up to undertake the roles traditionally undertaken by the Trial Steering Committee (TSC) and Data Monitoring and Ethics Committee (DMEC). This committee will comprise of independent members including a Chair and at least two other independent members including a statistician. The independent members of the committee will be allowed to see unblinded data, but this data would not be reported to the other members of the research team. The role of this committee will include the reviewing of all serious adverse events which are thought to be treatment related and unexpected. The committee will meet at least annually or more often as appropriate.

13.4 STUDY COLLABORATORS/PARTNERS

- Harry Campbell, Professor of Genetic Epidemiology and Public Health, is the Chief Investigator for RESPIRE. He is responsible for the overall conduct of RESPIRE projects.
- Kamran Siddiqi, Professor in Global Public Health at the University of York, is the UK PI for this study and the overall lead researcher for this study. He is responsible for ensuring that the trial is conducted in accordance with the approved protocol.
- John Norrie, Professor of Medical Statistics and Trial Methodology at the University of Edinburgh, is the statistician for this study. He is involved in the design of the study and will make sure that data collected is analysed using appropriate methods and is interpreted correctly.
- Steve Parrott, Reader in Health Economics at the University of York, is involved in the design of the health economic component and will conduct the cost analysis of the data collected.
- Rumana Huque, Bangladesh trial site PI, is responsible for the conduct of the trial at the Banglaesh site. She will provide training to site staff on the delivery of the protocol, maintain oversight of the Bangladesh trial site and take appropriate action to ensure good recruitment, and the quality and timeline of the data collection.
- Amina Khan, Pakistan trial site PI, is responsible for the conduct of the trial at this site. She will provide training to site staff on the delivery of the protocol, maintain oversight of the Bangladesh trial site and take appropriate action to ensure good recruitment, and the quality and timeline of the data collection. She will be responsible for retaining clear, accurate and complete records of all trial documentation. She will report on all safety issues and suspected breaches.
- Ai Keow Lim, RESPIRE Programmes and Trial Manager, is the study trial manager and is responsible for the management and organisation of the trial. She will prepare sponsor and ethics committee submissions and ensure that appropriate approvals

are obtained prior to the commencement of the study. She will work with the site PIs and coordinators to maintain oversight of the trial sites. She will submit proposed amendments to the protocol to the Sponsor and ethics committees.

- The Research Fellows/Trial Coordinators (to be recruited) will be responsible for identifying and recruiting potential research participants, supervise, monitor and coordinate the data collection, help maintain synergy between technical and support services at field sites, monitor field activities and report to site PI on identified issues solutions, supervise, coordinate and support field staff and organise field level meetings.
- Ms Samina Huque, Database Manager at ARK Foundation, is responsible for developing the data management plan and the management of the data collected. They will coordinate the design, build, test and validation of the case report forms and database. They will ensure timely returns of the data, entering the data onto the trial database and checking for inconsistencies and incomplete data. They will perform regular checks to validate and verify information on the database. They will work with RESPIRE Open Science, Data and Methodologies Platform Coordinator to ensure that data are transferred to UoE in safe and secure ways.
- Mr Saeed Ansari, Database Manager at The Initiative, is responsible for developing the data management plan and the management of the data collected. They will coordinate the design, build, test and validation of the case report forms and database. They will ensure timely returns of the data, entering the data onto the trial database and checking for inconsistencies and incomplete data. They will perform regular checks to validate and verify information on the database. They will work with RESPIRE Open Science, Data and Methodologies Platform Coordinator to ensure that data are transferred to UoE in safe and secure ways.

13 GOOD CLINICAL PRACTICE

13.1 ETHICAL CONDUCT

The study will be conducted in accordance with the principles of the International Conference on Harmonisation Tripartite Guideline for Good Clinical Practice (ICH GCP). In addition to the principles of the ethics committee(s)/Institutional Review Boards (IRBs) who have reviewed and approved this study.

Before the study can commence, all required approvals will be obtained, and any conditions of approvals will be met.

This study will receive approval from the University of Edinburgh’s Medical School Ethics Committee (EMREC) in addition to the Bangladesh Medical Research Council and the National Bioethics Committee-PMRC in Pakistan.

13.2 PRINCIPAL INVESTIGATOR RESPONSIBILITIES

The Principal Investigator is responsible for the overall conduct of the study at the site and compliance with the protocol and any protocol amendments. In accordance with the principles of ICH GCP, the following areas listed in this section are also the responsibility of the Principal Investigator. Responsibilities may be delegated to an appropriate member of study site staff.

13.2.1 Informed Consent

The Principal Investigator is responsible for ensuring informed consent is obtained before any protocol specific procedures are carried out. The decision of a participant to participate in clinical research is voluntary and should be based on a clear understanding of what is involved.

Participants must receive adequate oral and written information – appropriate Participant Information and Informed Consent Forms will be provided. The oral explanation to the participant will be performed by the Principal Investigator or qualified delegated person and must cover all the elements specified in the Participant Information Sheet and Consent Form.

The participant must be given every opportunity to clarify any points they do not understand and, if necessary, ask for more information. The participant must be given sufficient time to consider the information provided. It should be emphasised that the participant may withdraw their consent to participate at any time without loss of benefits to which they otherwise would be entitled.

The participant will be informed and agree to their medical records being inspected by regulatory authorities and representatives of the sponsor.

The Principal Investigator or delegated member of the trial team and the participant will sign and date the Informed Consent Form(s) to confirm that consent has been obtained. The participant will receive a copy of this document and a copy filed in the Principal Investigator Site File (ISF) and participant’s medical notes (if applicable).

13.2.2 Study Site Staff

The Principal Investigator must be familiar with the protocol and the study requirements. It is the Principal Investigator’s responsibility to ensure that all staff assisting with the study are adequately informed about the protocol and their trial related duties.

13.2.3 Data Recording

The Principal Investigator is responsible for the quality of the data recorded in the Case Report Form (CRF) at each Principal Investigator Site.

13.2.4 Principal Investigator Documentation

The Principal Investigator will ensure that the required documentation is available in local Investigator Site files ISFs.

13.2.5 GCP Training

All researchers will be encouraged to undertake GCP training in order to understand the principles of GCP. For participating in this trial, this will be a mandatory requirement for all researchers. GCP training status for all investigators should be indicated in their respective CVs.

13.3 DATA PROTECTION TRAINING

All University of Edinburgh employed researchers, students and study staff will complete the Data Protection Training through Learn.

13.4 INFORMATION SECURITY TRAINING

All University of Edinburgh employed researchers, students and study staff will complete the Information Security Essentials modules through Learn and will have read the minimum and required reading setting out ground rules to be complied with.

13.4.1 Confidentiality

All laboratory specimens, evaluation forms, reports, and other records must be identified in a manner designed to maintain participant confidentiality. All records must be kept in a secure storage area with limited access. Clinical information will not be released without the written permission of the participant. The Principal Investigator and study site staff involved with this study may not disclose or use for any purpose other than performance of the study, any data, record, or other unpublished information, which is confidential or identifiable, and has been disclosed to those individuals for the purpose of the study. Prior written agreement from the sponsor or its designee must be obtained for the disclosure of any said confidential information to other parties.

13.4.2 Data Protection

All Principal Investigators and study site staff involved with this study must comply with the requirements of the appropriate data protection legislation (including the European Union General Data Protection Regulation, the Data Protection Act 2018 in the UK and any relevant Data Protection laws in Bangladesh and Pakistan, respectively with regard to the collection, storage, processing, and disclosure of personal information.

Computers used to collate the data will have limited access measures via usernames and passwords.

Published results will not contain any personal data and be of a form where individuals are not identified, and re-identification is not likely to take place.

14 STUDY CONDUCT RESPONSIBILITIES

14.1 PROTOCOL AMENDMENTS

Any changes in research activity, except those necessary to remove an apparent, immediate hazard to the participant in the case of an urgent safety measure, must be reviewed and approved by the Chief Investigator.

Amendments will be submitted to the sponsor for review and authorisation before being submitted in writing to the appropriate ethics committees/IRBs, for approval prior to participants being enrolled into an amended protocol.

14.2 MANAGEMENT OF PROTOCOL NON-COMPLIANCE

Prospective protocol deviations, i.e., protocol waivers, will not be approved by the sponsor and therefore will not be implemented, except where necessary to eliminate an immediate hazard to study participants. If this necessitates a subsequent protocol amendment, this should be submitted to the appropriate ethics committees/IRBs for review and approval if appropriate.

Protocol deviations will be recorded in a protocol deviation log and logs will be submitted to the sponsor every 3 months. Each protocol violation will be reported to the sponsor within 3 days of becoming aware of the violation. All protocol deviation logs, and violation forms should be emailed to the sponsor at: QA@accord.scot

Deviations and violations are non-compliance events discovered after the event has occurred. Deviation logs will be maintained for each site in multi-centre studies. An alternative frequency of deviation log submission to the sponsors may be agreed in writing with the sponsors.

14.3 SERIOUS BREACH REQUIREMENTS

A serious breach is a breach which is likely to effect to a significant degree:

(a) the safety or physical or mental integrity of the participants of the trial; or

(b) the scientific value of the trial.

If a potential serious breach is identified by the Chief investigator, Principal Investigator or delegates, the sponsor (seriousbreach@accord.scot) must be notified within 24 hours. It is the responsibility of the sponsor to assess the impact of the breach on the scientific value of the trial, to determine whether the incident constitutes a serious breach and report to research ethics committees as necessary.

14.4 STUDY RECORD RETENTION

All study documentation will be kept for a minimum of 3 years from the protocol defined end of study point. When the minimum retention period has elapsed, study documentation will not be destroyed without permission from the sponsor.

14.5 END OF STUDY

The end of study is defined as the last participant's last visit.

The Investigators or the sponsor has the right at any time to terminate the study for clinical or administrative reasons.

The end of the study will be reported to the appropriate ethics committees/IRBs and sponsor within 90 days, (or within the specific timelines specified by the individual ethics committees/IRBs). If the study is terminated prematurely, the end of study will be reported to the relevant ethics committees/IRBs and sponsor within 15 days. The Investigators will inform participants of the premature study closure and ensure that the appropriate follow up is arranged for all participants involved.

End of study notification will be reported to the sponsor via email to researchgovernance@ed.ac.uk and to the relevant ethics committees who approved the study.

A summary report of the study will be provided to the appropriate ethics committee(s)/IRBs within 1 year of the end of the study (or within the specific timelines specified by the individual ethics committees/IRBs).

14.6 CONTINUATION OF TREATMENT FOLLOWING THE END OF STUDY

There are no plans to continue to provide the interventions (mTB-Tobacco and face-2-face behavioural counselling) beyond the trial. However, once policy makers have seen the trial results and have agreed to support one or both interventions, the investigators will support the respective TB programmes in implementing these interventions with the help of seeking implementation grants e.g., through Global Fund and TB Reach.

14.7 INSURANCE AND INDEMNITY

The sponsor is responsible for ensuring proper provision has been made for insurance or indemnity to cover their liability and the liability of the Chief Investigator and staff.

The following arrangements are in place to fulfil the sponsor responsibilities:

- The Protocol has been designed by the Chief Investigator and researchers employed by the University and collaborators. The University has insurance in place (which includes no-fault compensation) for negligent harm caused by poor protocol design by the Chief Investigator and researchers employed by the University.
- Sites participating in the study will be liable for clinical negligence and other negligent harm to individuals taking part in the study and covered by the duty of care owed to them by the sites concerned. The sponsor requires individual sites participating in the study to arrange for their own insurance or indemnity in respect of these liabilities.
- Sites in Bangladesh and Pakistan will be responsible for arranging their own indemnity or insurance for their participation in the study, as well as for compliance with local law applicable to their participation in the study.

15 STAKEHOLDER ENGAGEMENT

In Pakistan, we will engage with the TB programmes at both national and provincial levels and the tobacco control cell, including national and provincial focal points, from the start. In Bangladesh, we will engage with the National TB programme and the Directorate responsible for non-communicable diseases (including tobacco control) within the Ministry of Health. We will hold regular meetings with the respective focal persons in the afore-mentioned directorates and programmes and invite them to attend our steering committee meetings and dissemination events. The proposed study will strengthen our existing engagement with the respective TB control programmes in the two countries and help them in adopting the most cost-effective approaches to smoking cessation.

16 REPORTING, PUBLICATIONS AND NOTIFICATION OF RESULTS

16.1 CONFIDENTIALITY

The Parties each agree to use reasonable endeavours to keep confidential and not to publish or disclose in any way other than to those of its employees, students, directors, officers, advisors or representatives who have a need to know such information for the purposes of the Project:

- any Background IP of another Party identified as confidential at the time of disclosure; or
- any Results of another Party; or
- Joint Results;

(together the "**Confidential Information**")

without the consent of the Party owning or controlling such Confidential Information for a period of 5 years from the conclusion of the Project.

16.2 AUTHORSHIP POLICY

Any publication or other dissemination of the Results (or any part of them) by any of the Parties shall not occur until Edinburgh has published the Results of the Project in the primary publication (the "Primary Publication"). Authorship of the Primary Publication shall be in accordance with normal academic practice. Each Party shall be entitled to publish articles directly arising from its solely owned Results. Prior to the publication of articles directly arising from the work of more than one Party on the Project, each Party shall endeavour to circulate proposed publications at least 30 days in advance of submission for publication. All publications shall acknowledge the funding made available for the Project by the NIHR as required by the NIHR Award Terms. Each Party retains the right to request (such request not to be unreasonably refused) the delay of a publication in order to seek Intellectual Property protection for Results generated in the course of the Project if publication would reasonably prejudice such protection. Such delay shall not exceed 3 months, unless mutually agreed between the relevant Parties. Notification of the requirement for delay in submission for publication must be received by the publishing Party within 30 days after the receipt of the material by the other Party/Parties, failing which the publishing Party shall be free to assume that the other Party/Parties has no objection to the proposed publication.

These provisions clause shall survive termination or expiry of this Agreement for the period of 1 year.

No Party shall use the name or any trademark or logo of any other Party or the name of any of its staff or students in any press release or product advertising, or for any other commercial purpose, without the prior written consent of the Party(s).

Based on the study results, we will publish a range of high-quality materials and scientific papers from this trial. Therefore, it is hoped that all members of the team will engage with this aspect of the work, either by leading publications or through contributing to the writing process.

Opportunities to lead or participate in writing will be open to all members of the team who have made substantial contributions to the trial, whether through conceiving the work, securing funding, managing the project, conducting field work, generating, or interpreting data, or writing up findings.

Our publication process will be open and fair. This means that it will be transparent, so that all who wish to engage and/or contribute are aware of any opportunities or planned publications. However, authorship carries responsibilities, and all participants need to be aware of these. As per standard academic practice, authorship can only be rightly claimed if the ICMJE criteria are met. Where a manuscript or conference abstract is based on the methods and data of the trial, it may be appropriate to limit authorship to the members of the study team. However, after listing authors' names, a suffix will be added, with the words 'the RESPIRE Collaboration'. If the journal allows, all individually named authors who are members of RESPIRE should list "NIHR Global Health Research Unit on Respiratory Health (RESPIRE), The Usher Institute, University of Edinburgh, Edinburgh, UK" as one of their affiliations.

In order to promote inter-institutional collaboration within the trial, all manuscripts and conference abstracts will include authors from more than one institution, wherever possible. It is not expected that any one person within the team takes responsibility for policing adherence to the above authorship criteria. All members are expected to adhere to their own professional and/or academic codes of conduct, in line with the criteria above.

It is also important that the team members do not breach other ethical standards with respect to academic publishing. These include declaration of conflicts of interest, protecting confidentiality/anonymity, rules on gift authorship and ghost authorship, salami (segmented) publication and duplicate publication. (See http://www.wame.org for further information).

17 REFERENCES

1 WHO | Global tuberculosis report 2018. 2019; published online March 3. https://www.who.int/tb/publications/global_report/en/ (accessed Aug 29, 2019).

2 Zellweger J-P, Cattamanchi A, Sotgiu G. Tobacco and tuberculosis: could we improve tuberculosis outcomes by helping patients to stop smoking? European Respiratory Journal. 2015; **45**: 583–5.

3 Lin HH, Ezzati M, Murray M. Tobacco smoke, indoor air pollution and tuberculosis: a systematic review and meta-analysis. *PLoS Med* 2007; **4**: e20.

4 Lönnroth K, Castro KG, Chakaya JM, *et al.* Tuberculosis control and elimination 2010–50: cure, care, and social development. *Lancet* 2010; **375**: 1814–29.

5 Slama K, Chiang CY, Enarson DA, *et al.* Tobacco and tuberculosis: a qualitative systematic review and meta-analysis. *International Journal of Tuberculosis and Lung Disease* 2007; **11**: 1049–61.

6 Leung CC, Yew WW, Chan CK, *et al.* Smoking adversely affects treatment response, outcome and relapse in tuberculosis. *Eur Respir J* 2015; **45**: 738–45.

7 Yen YF, Yen MY, Lin YS, *et al.* Smoking increases risk of recurrence after successful anti-tuberculosis treatment: a population-based study. *Int J Tuberc Lung Dis* 2014; **18**: 492–8.

8 Marshall A-M, Barua D, Mitchell A, *et al.* Smoking prevalence among tuberculosis patients: A crosssectional study in Bangladesh and Pakistan. *Tob Induc Dis* 2020; **18**: 70.

9 Warsi S, Elsey H, Boeckmann M, *et al.* Using behaviour change theory to train health workers on tobacco cessation support for tuberculosis patients: a mixed-methods study in Bangladesh, Nepal and Pakistan. *BMC Health Serv Res* 2019; **19**: 71.

10 Siddiqi K, Khan A, Ahmad M, *et al.* Action to Stop Smoking In Suspected Tuberculosis (ASSIST) in Pakistan: a Cluster Randomized, Controlled Trial. *Ann Intern Med* 2013; **158**: 667–75.

11 Integrating Tobacco Control into Tuberculosis and HIV Responses. UNDP. https://www.undp.org/content/undp/en/home/librarypage/hiv-aids/integrating-tobacco-control-into-tuberculosis-and-hiv-responses.html (accessed Sept 26, 2019).

12 Dogar OF, Keding A, Gabe R, *et al.* Multicentre Randomised Double-Blind Placebo Controlled Trial of Cytisine for Smoking Cessation in Smokers with Tuberculosis. *The Lancet Global Health* 2020; published online June 24. http://eprints.whiterose.ac.uk/162580/ (accessed Aug 2, 2020).

13 Siddiqi K, Keding A, Marshall A-M, *et al.* Effect of quitting smoking on health outcomes during treatment for tuberculosis: secondary analysis of the TB & Tobacco Trial. *Thorax* 2021; published online July 16. DOI:10.1136/thoraxjnl-2020-215926.

14 Dogar O, Elsey H, Khanal S, Siddiqi K. Challenges of Integrating Tobacco Cessation Interventions in TB Programmes: Case Studies from Nepal and Pakistan. *J Smok Cessat* 2016; **11**: 108–15.

15 Organization WH, Others. Be he@ lthy, be mobile: a handbook on how to implement mTB-Tobacco. 2019. https://apps.who.int/iris/bitstream/handle/10665/325243/9789241515900-eng.pdf.

16 West R, Hajek P, Stead L, Stapleton J. Outcome criteria in smoking cessation trials: proposal for a common standard. Addiction. 2005; **100**: 299–303.

17 Dogar O, Keding A, Gabe R, Marshall AM, Huque R, Barua D, Fatima R, Khan A, Zahid R, Mansoor S, Kotz D, Boeckmann M, Elsey H, Kralikova E, Parrott S, Li J, Readshaw A, Sheikh A, Siddiqi K; TB and Tobacco Consortium. Cytisine for smoking cessation in patients with tuberculosis: a multicentre, randomised, double-blind, placebo-controlled phase 3 trial. Lancet Glob Health. 2020 Nov;8(11):e1408-e1417. doi: 10.1016/S2214-109X(20)30312-0. PMID: 33069301

Study protocol v6.0– latest version

Global Health Research

Non-CTIMP Study Protocol

A mHealth intervention (mTB-Tobacco) for smoking cessation in people with tuberculosis: a two-stage adaptive design, randomised trial

Quit4TB Trial

Sponsor	The University of Edinburgh ACCORD The Queen's Medical Research Institute 47 Little France Crescent Edinburgh EH16 4TJ
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Funder	NIHR Global Health Research Unit on Respiratory Health (RESPIRE)
Funding Reference Number	NIHR132826
Chief Investigator	Professor Harry Campbell
Sponsor number	AC22139
Ethics Number	UK: EMREC-RESPIRE-23-01 Bangladesh: BMRC/NREC/2022-2025/176 Pakistan: NIH NBC No.4-87/NBC-929/23/1873
Trial registration number	ISRCTN86971818 https://www.isrctn.com/ISRCTN86971818
Version Number and Date	V6.0 7th March 2025

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FUNDED BY NIHR | National Institute for Health and Care Research
UK International Development
Partnership | Progress | Prosperity

LIST OF ABBREVIATIONS

ACCORD	Academic and Clinical Central Office for Research & Development
CI	Chief Investigator
PI	Principal Investigator
CRF	Case Report Form
GCP	Good Clinical Practice
ICH	International Conference on Harmonisation
LMIC	Low- and Middle-Income Country
DOTS	Direct Observation Treatment Strategy
QA	Quality Assurance
SOP	Standard Operating Procedure
TB	Tuberculosis

18 INTRODUCTION

18.1 BACKGROUND

Tuberculosis (TB) affects >10 million people every year and approx. 1.5 million die as a result.[1] While its incidence has declined in many parts of the world, the global TB pandemic continues to be fuelled by comorbid conditions (e.g. HIV, diabetes) and risk factors (e.g. smoking,[2] indoor air pollution[3]). Among these, smoking alone is responsible for 16% of the total TB disease-burden.[4] Smoking increases the chances of acquiring TB infection and TB disease by two- and three-folds, respectively;[5] continued smoking by those with active TB disease worsens their clinical outcomes.[3] Compared to non-smokers, TB patients who smoke are more likely to have severe forms of clinical presentation, poor bacteriological and clinical responses to treatment(s), and high rates of treatment failure.[6] Smoking doubles the risk of TB-related deaths[5] and, in survivors, TB recurrence.[7]

Smoking prevalence in TB patients is higher than the general population,[8] but the vast majority of TB patients are neither routinely asked about their smoking status nor advised to quit.[9] The strong association between smoking and TB has led to increasing recognition of the need for evidence-based smoking cessation approaches[10] and increasing policy support to help TB patients quit smoking.[11]

18.2 RATIONALE FOR STUDY

In a series of randomised-controlled trials (RCT)[10,12] in Bangladesh and Pakistan, our team has developed strong evidence to support face-to-face behavioural interventions achieving high quit rates.[10] We have also demonstrated that those who quit smoking show better clinical outcomes.[13] However, for several health system barriers[14] (cost, reach, sustainability) no TB high-burden country has so far integrated face-to-face smoking cessation within TB services. Recognising the challenges of integrating and scaling up face-to-face interventions, we have worked with WHO to develop a mHealth smoking cessation package (mTB-Tobacco) that can be delivered as SMS messages via mobile phones in TB patients.[15] While delivering mHealth package will cost less than face-to-face behavioural interventions, it is unknown if this package is as effective and therefore more cost-effective than face-to-face behavioural intervention. We have also consulted with TB programme managers in Bangladesh and Pakistan, colleagues working in TB DOTS Partnership and with the International Union Against TB and Lung Diseases, an evaluation of a mHealth package to achieve smoking cessation in TB patients has been identified as a priority.

19 STUDY OBJECTIVES

19.1 OBJECTIVES

19.1.1 Primary Objective

In people with TB who smoke daily, we will assess the effectiveness and cost effectiveness of mTB-Tobacco in achieving continuous abstinence for at least six months.

19.1.2 Secondary Objectives

In people with TB who smoke daily, we will assess the effectiveness and cost effectiveness of mTB-Tobacco in enhancing TB treatment adherence and improving clinical outcomes.

19.2 ENDPOINTS

19.2.1 Primary Endpoint

The primary endpoint will be biochemically verified continuous abstinence at 6 months post randomisation.[16] Abstinence is defined as self-report of not having used more than 5 cigarettes, bidis, water pipe sessions since the quit date, verified biochemically by a breath carbon monoxide (CO) reading of less than 10 ppm at month 6.

In case of concomitant smokeless tobacco use, biochemical verification will be carried out using Accutest® NicAlertTM strips to detect cotinine (a nicotine metabolite) level in urine samples. Reading of the strip will indicate a cotinine level (0-6), where level <3 (i.e., equivalent to 100-200ng/mL cotinine) will be considered tobacco abstinence.

When a patient self-reports abstinence with an elevated CO level of >10ppm (and cotinine levels in concomitant users indicating active tobacco use), the biochemical verification will supersede the self-report and the patient will be defined as a tobacco user.

19.2.2 Secondary Endpoints

Secondary outcomes will include:

10. Point abstinence, defined as a self-report of not using tobacco in the previous 7 days, assessed at week 9 and month 6.
11. Adherence to TB Treatment: All registered TB patients' medication logs (for anti-TB medication) are recorded on the 'Treatment Support Card'; a copy of this card would be requested from the TB paramedic by the RA and attached with the patient CRF.
12. TB Programme outcomes: the proportion of treatment success (including cured and completed treatment), treatment failure, defaulted and died will be recorded from TB register (TB03) at month 6. The definitions of these outcomes are as follows:
13. Cured: 'A patient who was initially smear-positive and who was smear negative in the last month of treatment (at month 6) and on at least one previous occasion'
14. Completed treatment: 'A patient who completed treatment (at month 6) but did not meet the criteria for cure or failure.'
15. Treatment failure: 'A patient who was initially smear-positive and who remained smear-positive at month 6 or later during treatment'
16. Defaulted: 'A patient whose treatment was interrupted for 2 consecutive months or more'
17. Died: 'A patient who died from any cause during treatment'
18. Relapse: 'A patient who was previously treated for TB, was declared cured or treatment completed at the end of his most recent course of treatment, and is now

diagnosed with a recurrent episode of TB (either a true relapse or a new episode of TB caused by reinfection)'

20 STUDY DESIGN

Using an adaptive design, we will conduct a multi-centre, cluster randomised, controlled trial consisting of four phases. Phase 1 involves consultation with Patients and Public Involvement (PPI) group about the study processes. PPI members will also review and provide feedback on the participant facing materials. Phase 2 is the pilot study. In Phase 3 **(superiority trial)**, which will last for 12 months (6 months recruitments and 6 months follow ups), we will compare mTB-Tobacco (intervention A) with usual care (control). In Phase 4 **(non-inferiority trial),** which will last for another 12 months (6 months recruitment and 6 months follow ups), we will compare mTB-Tobacco (intervention A) with face-to-face behavioural support (intervention B). See QUIT4TB Trial flow diagram 1 below for details.

21 STUDY POPULATION

21.1 STUDY LOCATION

We will conduct the trial in two countries with high TB and tobacco-disease burden:

	TB (per 100,000)	TB cases (n)	TB deaths (n)	Tobacco prevalence
Bangladesh	221	357,000	47,000	35%
Pakistan	265	562,000	44,000	20%

We will recruit TB health facilities (clusters) in the public sector from Dhaka division in Bangladesh and Punjab province in Pakistan.

21.2 SAMPLE SIZE

The study requires a total of 2,716 smokers with TB (~43 recruits from **63 health facilities** [the clusters]) assuming that 10% do not provide primary outcome data. This sample also includes the first 16 participants taking part in the pilot study in case the intervention needed major refinement and we are unable to use their data. Phase 3 comparing A with C, would have 90% power at 5% significance level to compare 18 clusters randomised to A with 9 clusters randomised to C, assuming proportion achieving abstinence in usual care was 8% and using the mTB-Tobacco was 18% abstinence at 6 months. If A was superior to C, Phase 4 would then continue, and would have 90% power at a 1-sided 2.5% level to establish non-inferiority of mobile to face-to-face, comparing 18 clusters in each of A and B assuming face-to-face achieved 18% abstinence at six months and the non-inferiority margin was 8%. Assuming that just 2% would give up with no intervention at all, the established treatment effect of face-to-face over natural cessation over 6 months would then be 16% (18%-2%). The non-inferiority margin of 8% equates to the mobile intervention preserving at least 50% (8/16) of the established treatment effect of the face-to-face intervention. We will check above sample size assumptions mid-way through the recruitment period and adjust the sample size accordingly.

21.3 INCLUSION CRITERIA

21.3.1 Clusters (health facilities) inclusion criteria

All TB health facilities with a minimum turnover of 50 new TB patients a month will be eligible. The included sites must be functioning, and designated TB diagnostic centres approved by the National /Provincial TB control Programme (NTP/PTP). These may include primary, secondary, or tertiary care health centres (HCs), generally known as the district/or tehsil headquarter hospitals in Pakistan. In Bangladesh, health facilities will be the DOTS corners at Upazila (sub-district) Health Complex and Urban DOTS corners. In addition, the following conditions must be met:

- The facility must be able to nominate a site focal person (an NTP personnel e.g., the district TB coordinator, or the responsible TB clinician at the site) who assumes responsibility for the proper conduct of the trial at their site.
- The HC should have available an adequate number of qualified staff and adequate facilities for the foreseen duration of the trial to conduct the trial properly and safely.
- The HC should be willing to join the trial and provide required administrative support.

21.3.2 Participants inclusion criteria

Patients will be considered eligible for enrolment in this trial if they fulfil all the following inclusion criteria and none of the exclusion criteria (see 4.4.2).

- Age at least 15 years (counted as adult TB patients in Bangladesh and Pakistan)
- Willing and able to provide written informed consent
- Diagnosed with drug sensitive PTB (smear positive or negative) in the last four weeks
- Currently smokes tobacco on a daily basis or have only stopped or reduced smoking (less than daily) since diagnosed with TB
- Willing to quit tobacco use
- Access to personal mobile phone

21.4 EXCLUSION CRITERIA

21.4.1 Clusters (health facilities) exclusion criteria

Facilities not meeting the inclusion criteria will not be eligible for the trial.

21.4.2 Participants exclusion criteria

Patients anticipated to have adverse effects either due to the study treatment or research burden, will be excluded. These will include those who are/have:

- Less than 15 years of age
- Retreatment TB, MDR TB, Miliary or Extrapulmonary TB
- Currently using any pharmacotherapy for tobacco dependence
- Unwilling or unable to provide written informed consent

22 PARTICIPANT SELECTION AND ENROLMENT

22.1 IDENTIFYING PARTICIPANTS

Doctors, paramedic/nurses, Health Workers (HW), other clinical staff and laboratory technicians engaged in TB care and management at the HC will identify TB patients. It is anticipated that the bulk of the patients will be identified through the TB registers kept by the TB paramedics (aka TB DOTS facilitators), responsible for recording, reporting, and dispensing anti-TB medications to the patients. Once a patient is identified, the resident Research Assistant (RA) will complete the Screening Form for assessing eligibility.

22.2 CONSENTING PARTICIPANTS

Eligible patients will be given verbal and written information about the trial and 24 hours to consider participation. They will have the opportunity to discuss the trial fully with the RA (and the HW/TB paramedic if patient requests). Each participant will be informed of the aims, methods, anticipated benefits, potential hazards, and discomforts of the study, both through the Patient Information Sheet and verbally. We will be using the same procedures of obtaining consent as in the previous trial[17] sponsored by the University of York. The research assistant (RA) will be trained to obtain informed consent from the participants. There is unlikely to be information loss unless it is done in a rush. The main points will be summarised to bullet points for the RA to read to the participants, which will reduce the scope of information loss to a minimum. The RA will also check orally with the participants if they have understood the key information. The patient's right to decline participation and the right to withdraw at any time, without the need to give an explanation and without it being detrimental to their overall treatment will be clearly stated. Each participant will be reimbursed for their travel expenses based on the distance travelled (400PKR/visit (fixed amount); up to 150BDT/visit, based on distance travelled; £1.10/visit) for their week-9 and 6-month follow-up visits. Written consent will be obtained from those interested by the RA – trained (by the local PIs and Research Fellows/Trial Coordinators) on the appropriate information and consent procedures for the TB patients (prior to the start of patient enrolment in study). All participants agreeing to participate will be required to sign (or place a thumbprint using indelible ink) the consent form alongside their National Identity Card number (if any), address and mobile telephone number.

22.2.1 All consent forms will be stored in accordance with local requirements. A copy of the consent form will be given to the patient and the original signed copy kept in a locked cabinet in the respective research offices. The Research Fellows/Trial Coordinators will bring the original copy to the corresponding country coordinating centre for central monitoring purposes. Withdrawal of Study Participants

Participants are free to withdraw from the study at any point or a participant can be withdrawn by the Principal Investigator. The Principal Investigator (PI) may withdraw a participant if it is considered in the participant's best interest not to continue the study e.g. if participant's health has deteriorated and their continued participation may undermined their health further. In rare situation that the PI decides to withdraw a participant, they will check with other clinicians. If withdrawal occurs, the primary reason for withdrawal will be documented in the participant's case report form, if possible. If a participant decides to withdraw, they will be informed that:

a. They will be withdrawn from all aspects of the trial but continued use of data (and samples) collected up to that point.

To safeguard rights, the minimum personally identifiable information possible will be collected.

Detail reasons and procedures for a study participant stopping early i.e., “stopping rules” and “discontinuation criteria”

23 RANDOMISATION AND ALLOCATION PROCEDURES

Since the interventions will be delivered at a group level, this trial is a cRCT where health facilities will be randomly allocated to the different trial arms. The health facilities (clusters) would be randomised 2:2:1 to mTB-Tobacco (intervention A), face-to-face (intervention B), or standard care (Control). An independent statistician based at University of Edinburgh who will be blinded to centres, will identify and use computer-generated random-number lists to generate the allocation sequence. The allocation will be based on using minimisation to ensure balance across the groups on the average number of TB patients seen per month and geographical location (Bangladesh and Pakistan).

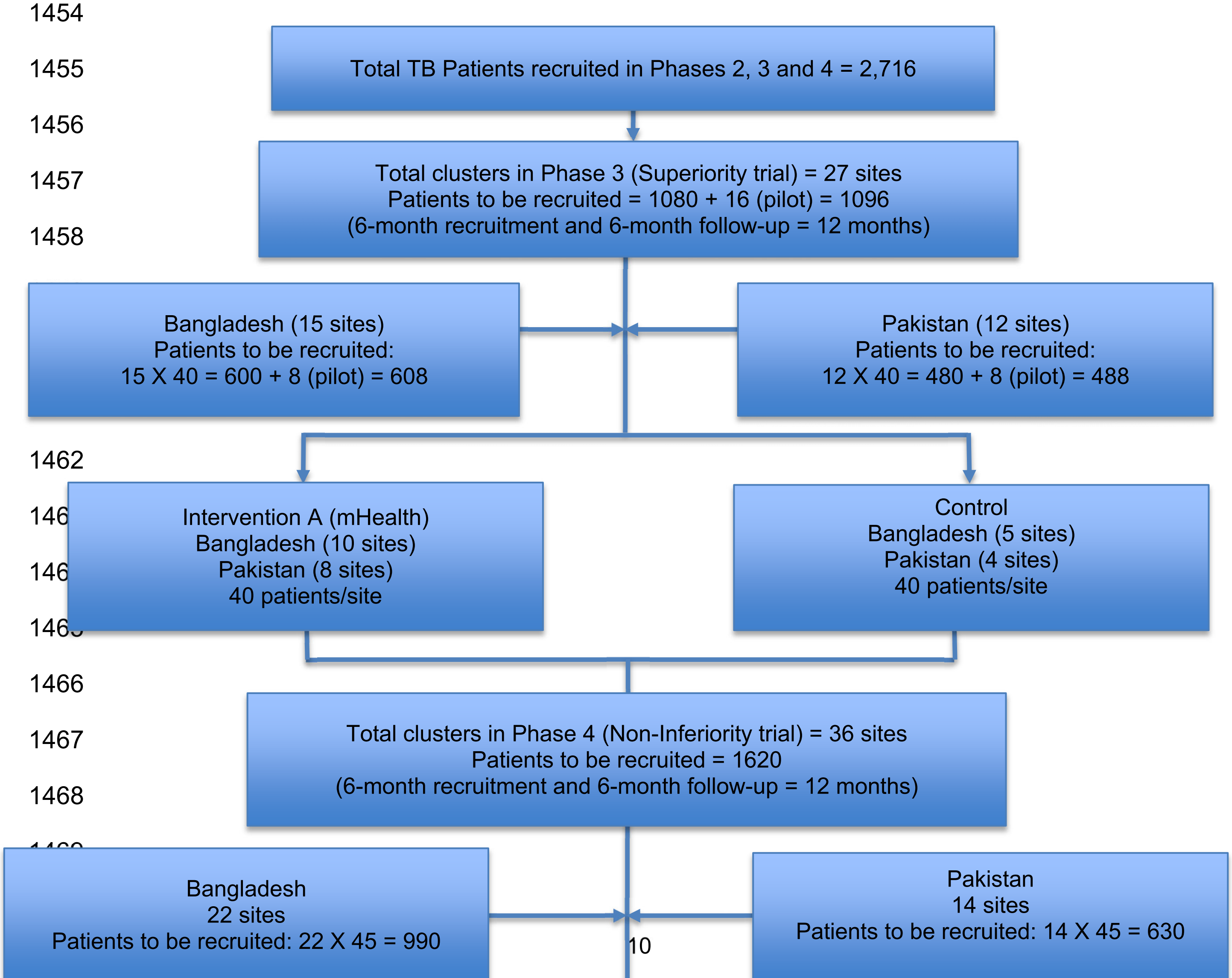

Diagram 1: Quit 4 TB Trial flow diagram

24 INTERVENTIONS

24.1 mTB-TOBACCO (INTERVENTION A)

The mTB-Tobacco programme (intervention A) is designed to send short message service (SMS) content with two distinct areas of intervention. The first tries to instil behaviour change in patients to quit the habit of tobacco use and the second includes a combination of supportive motivational and informative messages through the period of TB treatment. The overall outcome being that a person is able to both successfully quit tobacco consumption and complete TB treatment.

A generic intervention has already been developed by the investigators in collaboration with the WHO mHealth team, available here: https://www.who.int/publications/i/item/a-handbook-on-how-to-implement-mtb-tobacco. This resource provides guidance for national TB control programmes and organisations responsible for delivering TB control in adapting and implementing a bespoke mTB-Tobacco programme. The resource includes a set of SMS messages to inform TB patients about the hazards of tobacco use and encourage them to quit. Furthermore, these messages are also designed to enhance treatment adherence, increase healthy behaviours, and reduce potentially harmful behaviours in TB patients in order to improve their health outcomes.

To make the intervention ready for the trial, the research team will set up a reference group (TB and tobacco experts, TB programme managers and clinicians) to support, and inform the adaptation of mTB-Tobacco intervention. The group will first select a set of SMS messages (from the abovementioned resource) and draft their transmission schedule (the number of messages, their frequency and time intervals). Once a plan of the mTB-Tobacco programme has been developed, it will go through iterative stages of pre-testing, piloting, and refinement.

In phase 1, we will organise PPI forum meetings to ask them to review our patient facing materials including SMS messages for comprehension, language, and tone. The PPI group will comprise of six-eight members including TB patients, carers and lay members of public, both in Bangladesh and Pakistan. Messages that are considered as unclear or inappropriate for the target group would be rewritten.

Once the messages have been reviewed by the PPI group, the intervention will be tested (as part of the pilot study) with a total of 16 patients (8 each in Bangladesh and Pakistan). The pilot will be evaluated using two sets of data: users' self-reported experiences and their real-time engagement with the programme. Users' experiences can be collected through face-to-face/phone-based interactions. Key questions would focus on users' experience of taking part in the mTB-Tobacco programme; the clarity, quantity, timing and frequency of messages; what was good about the programme and what was not; completion or non-completion the programme; and any effect on their attitudes or target behaviour(s). Users' real-time engagement will be evaluated using computer records from the programme used for launching mTB-Tobacco. The pilot study is embedded in the superiority trial (phase 3). The pilot participant data will be treated with the same processes and governance as the main trial participant data.

Once mTB-Tobacco is piloted, but before continuing to superiority trial, the programme will go through a refinement process (a study pause) that will address issues highlighted during the pilot and make adjustments to the intervention, if needed.

At the end of above exercise, the intervention will be ready for the superiority trial and will comprise of: (a) a set of SMS messages translated in Bangla and Urdu, respectively; (b) a schedule of transmission; (c) an automated digital SMS messaging platform ready to send these messages as per schedule; and (d) a dedicated team to supervise and monitor the transmission of these messages. The SMS messaging will only store unique ID number for the trial. No other personal or identifiable information will be stored. Each participant in the intervention group will receive a total of 134 SMS messages over a period of 6 months. In the first two months, the frequency of messages would be 4 to 5 messages per day, in next two month the frequency would reduce to 1 to 2 messages per day and in the last two months there will be 1 message sent per week. For phase 3, in addition to mHealth smoking cessation package, this group will also receive education leaflets from TB health professionals.

We are using REDCap for managing the data for this trial. The mHealth app will be linked to REDCap. At the screening stage, participants who meet all inclusion criteria and do not meet any exclusion criteria will be eligible to take part in the study. They will be given the participant information sheet and consent form. If they consent to take part in the study, their demographic details (including active mobile number) will be recorded on the REDCap database. If the participant is marked as belonging to the intervention arm, then his/her contact details/mobile number will be captured from the database by the mobile SMS service to send mHealth messages as per the schedule. The participants belonging to the intervention arm will start receiving the SMS messages, whereas those belonging to control arm will not receive any SMS.

The mobile SMS service is provided by a third-party service. A contract of services for the development of the mTB-Tobacco program application has been signed between The Initiative and Biter Tech (a third-party). Private Limited. Section A General Terms & Conditions Point 7 of the sub-contract (highlighted in yellow) mentions that "The Consultant is not allowed to keep or sell any project related data. The Consultant agrees to keep confidentiality of project data, and, to cause its employees and agents to keep confidentiality of any data available to them. The consultant will not access or check participant's personal data unless it is necessary in connection with the services provided under this Agreement, or otherwise consented to, in writing.”

24.2 FACE-2-FACE BEHAVIOURAL SUPPORT (INTERVENTION B)

This consists of an adapted version of a pre-developed and proven effective intervention to help people to quit smoking and smokeless tobacco use.[10] This consists of two face-to-face sessions delivered at day 0 and day 5 (+2) and which last 10 and 5 minutes respectively. The sessions will be structured using an educational flipbook; the session on day 0 will be aimed at encouraging tobacco users to see themselves as non-users and set a plan for a quit date five days later; with a session on the quit date (day 5) to review progress. Further encouragement and support (if needed) will be offered at a subsequent visit at week 5.

24.3 STANDARDISED USUAL CARE (INTERVENTION C)

The TB health professionals will be provided with patient education leaflets to be given out to all patients allocated to the respective trial arm. Leaflets for patients and others in the house contain information on the harmful effects of tobacco and advice on stopping its use.

25 STUDY ASSESSMENTS

25.1 STUDY ASSESSMENTS

All patients will have their follow-ups flagged up (in the trial database that can be printed off) to avoid attrition, both at the recruiting site and the respective country coordinating centre. The patients will have assessments on day 0, week 9 and month 6 time-points correspond with routine TB clinic visits. At the end of Phase 3 Superiority Trial, 20% of the participants in the mTB-Tobacco intervention groups in each country will be asked to complete a short Technology Acceptable Model (TAM) questionnaire. At the end of Phase 4 Non-inferiority Trial, all participants in the mTB-Tobacco intervention groups will be asked to complete the TAM questionnaire. The assessment schedule is tabled below.

Assessment	Screening	Day 0 baseline	Week 9	Month 6
Assessment of Eligibility Criteria	☒			
Written informed consent	☒			
Face to face / phone-based interaction*			☒	
Self-reported tobacco use/abstinence status		☒	☒	☒
CO measurement				☒
Socio-Demographic information		☒		
Tobacco use and quit history		☒		
Nicotine dependency		☒	☒	☒
Economic outcomes		☒		☒
Process outcomes (Dose delivered and fidelity)			☒	☒
Medication (TB) compliance			☒	☒
Adverse events review			☒	☒
TB outcomes				☒

TAM questionnaire (30% of participants in the mTB-Tobacco groups)**				☒

* This applies to Phase 2 pilot participants only.

** This applies to Phases 3 and 4 MTB-Tobacco groups only.

The trial assessment schedule will outline visit dates (with windows) necessary for data collection, but the patient may be seen more frequently for clinical care as needed.

During the treatment period, if patients are unable to attend on the day, every effort should be made to complete the visit within 2 days of the scheduled date. If a scheduled visit is missed without notice, then the RA should endeavour to contact the patient by phone or by home visit.

During the follow-up phase, scheduled assessments should be carried out no more than 5 days before the scheduled visit. HCs may choose to re-schedule visits to allow for public holidays or other unavoidable circumstances that affect the scheduled visit date, but the re-scheduled visit should be no more than 5 days from the originally scheduled visit date. Patients will also be given a card with the contact details for the trial RA and the clinical TB care team at their site.

If a patient is more than 5 days late for a scheduled study visit, an additional visit will be performed as soon as possible, including the appropriate assessments that were specified in the trial schedule for the visit that was missed.

20% of the total patients in the mTB-Tobacco intervention groups will be asked to complete a short Technology Acceptance Model (TAM) questionnaire over the phone with a research assistant.

Country	Phase 3 mTb-Tobacco group	Phase 4 mTb-Tobacco group
Bangladesh	10 sites X 40 participants per site X 20% = 80 participants	11 sites X 45 participants per site = 495 participants
Pakistan	8 sites X 40 participants per site X 20% = 64 participants	7 sites X 45 participants per site = 315 participants

The questionnaire is used to measure a user's perceived usefulness and perceived ease of use of using mTB-Tobacco to quit smoking and manage their TB. The questionnaire asks users to rate statements about the technology's usefulness, ease of use and how easy they think it will be to learn and use, on a scale ranging from strongly disagree to strongly agree. It will take about 5 to 10 minutes to complete the questionnaire.

26 DATA COLLECTION

This is a cRCT collecting quantitative data only. Following types of data will be collected:

- Personal identifiable data – this will include name, address, mobile telephone number and the national identity card number (if any) and will be stored together with the consent forms separate from other sources of data.

- Socio-demographic data – this will be collected only once at baseline using the Case Report Forms.
- Self-reported data on tobacco use behaviour (past and present) including quitting or staying abstinent, nicotine dependence and withdrawal symptoms – this will be collected only once at baseline using the Case Report Forms.
- We will also collect economic data which will include cost of the delivery of intervention collected from various sources (see economic analysis subsection), health service use and quality of life data collected through Case Report Forms.
- Process data will be collected to assess the fidelity to the delivery and receipt of SMS and face-2-face messages as part of mTB-Tobacco and behavioural support interventions, respectively. mTB-Tobacco fidelity data will be collected from the digital platform delivering messages and behavioural support fidelity data will be collected using the Case Report Form.
- Biochemical verification using CO monitors will be conducted for all those who report abstinence from tobacco use. This will be collected at the final follow using the Case Report Forms.
- TB medication adherence and TB outcomes data will be collected from routine TB data collected and recorded in TB registers, but this will also be validated through the Case Report Forms.
- TAM questionnaire comprises of 12 statements each on a five-point Likert scale ranging from strongly disagree to strongly agree.

The data collection will consist of responses to a set of structured questions in the Case Report Forms which will be completed and facilitated by the field team. Tablets will be used to enter data directly, which will appear in the database in real time. The data entered will synchronise with the database at the University of Edinburgh on a secured server. All the de-identified data will be transferred to the statistician in Edinburgh who will conduct the analysis using the statistical software packages: STATA and R.

Overall data quality will be ensured through training and supervision. Data entry validation will occur by double-checking a random sample of the data in the field. Moreover, quantitative data once entered will also be peer-reviewed by the Edinburgh-based statistician. We have used these processes successfully in previous studies in Pakistan and Bangladesh.

26.1 SOURCE DATA DOCUMENTATION

The following data should be verifiable from source documents:

- All signed consent forms
- Dates of visits
- Eligibility and baseline values for all participants
- All clinical endpoints (including TB related to the trial)
- Routine patient clinical and laboratory data
- Medication compliance
- Collection of personal data

- Written notes for face to face / phone-based interactions with Phase 2 pilot participants at week 9 (will be used to inform the protocol for Phases 3 and 4 of the study)

All information collected during the course of the study will be kept strictly confidential. The researchers will also comply with all aspects of the UK General Data Protection Regulation (GDPR), the Data Protection Act 2018, and the Privacy and Electronic Communications (EC Directive) Regulations 2003, and operationally this will include:

Consent from patients to record personal details including name, date of birth, and address.

Appropriate storage, restricted access, and disposal arrangements for patients' personal details.

Identifiable information will be collected on the consent form when he or she is consented into the study, but all other data transferred to or from the country coordinating centre will be coded with a Trial number and will not include patients' identifiers.

If a patient withdraws consent from further study participation, no further assessments will be conducted but their data already collected will remain on file and will be included in the final study analysis.

Data will be securely archived by University of Edinburgh for a minimum of 10 years. Following authorisation from the sponsor, arrangements for confidential destruction will then be made.

26.2 CASE REPORT FORMS

A validated case report form will be used to collect all participants' data at the baseline and follow ups. This will be captured digitally using an app on a password-protected tablet computer.

27 DATA MANAGEMENT

A trial specific data management plan (included in the Manual of Procedures (MoP)) agreed by the Chief Investigator, trial unit and other study investigators will be drafted to provide detailed instructions and guidance relevant to database set up, data entry, validation, review, query generation and resolution, quality control processes involving data access and transfer of data to the University of Edinburgh at the end of the study and archiving.

Participants' name and personal identifiers will be removed. All information will be coded and de-identified. The information we have collected in paper copies will be stored under lock and key cabinet at

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Road # 109,
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Bangladesh

The electronic data will be stored in a secured computer which can only be accessed with a secure password. Only the researchers will have access to the data.

27.1 DIRECT ACCESS TO SOURCE DATA

A statement of permission to access source data by study staff and for regulatory and audit purposes will be included within the patient consent form with explicit explanation as part of the consent process and Participant Information Sheet. Once pseudonymised, data will only be shared between respective country coordinating centres and the University of Edinburgh. Once the team has completed the analysis and published all intended scientific journals, it will be made available for other researchers.

In principle, pseudonymised data will be made available for meta-analysis and where requested by other authorised researchers and journals for publication purposes. Requests by other researchers for access to pseudonymised data archived on Edinburgh DataVault will be reviewed by the Principal Investigators. Edinburgh DataShare is an open access research data repository.

The Investigator(s)/Institutions will permit monitoring, and REC review (as applicable) and provide direct access to source data and documents.

Study data will be recorded in a number of files for both the administration of the study and collection of patient data.

All data will be completely pseudonymised for purposes of analysis and any subsequent reports or publications. For the purposes of ongoing data management, once randomised, individual patients will only be identified by Trial numbers, CRFs will be designed using ReDCAP software. The CRFs will be available as an online eCRF using a web-based interface.

27.1.1 Data entry

The data collected will be entered into a secure web-based interface, specifically developed for this study. Confirmation of the data received by the trial's unit will be given to the country trial manager.

All data will be stored and transferred following Health Insurance Portability and Accountability Act (HIPAA). HIPAA standards either comply with or exceed data protection regulations in operation in the two study countries, Bangladesh and Pakistan. Therefore, adherence to HIPAA ensures that sufficient data management standards are met for this international study. The staff involved in the trial (both at the sites and country coordinating centres) will receive training on data protection. The staff will be monitored to ensure compliance with privacy standards. Data will be checked according to procedures detailed in the trial specific Data Management Plan.

27.1.2 Data storage

Each site will hold data according to the Data Protection Act 2018 and data will be collated in CRFs identified by a unique identification number (i.e. the Trial number) only. For example, the Trial Screening Forms will contain a number to identify the site and a screening number (which will be the patient TB registration number- see section 3.3.1 on screening procedures for details) unique to each patient at that site, and once randomised, follow up forms will contain a Trial number unique to each patient in the trial. An Intervention Allocation Log at the coordinating centres will list the TB registration number of the patient against allocated Trial number for those randomised. The trial database will maintain a list of all TB registration numbers and Trial numbers for all trial patients at each site for the entire trial.

All study files will be stored in accordance with GCP guidelines. Electronic data will be password protected and stored on a secure server by the trial's unit. Only participants' identification numbers (site number, TB Registration number, Trial number) will be used to uniquely identify patients on the online electronic eCRFs.

Study documents (paper and electronic) at the participating sites, the country coordinating centres and the University of Edinburgh will be retained in a secure (locked when not in use) location, during and after the trial has finished. All personal data including identifying information will be stored in a secure data-storing facility and secure servers for a minimum of 10 years following the end of the study at:

The Initiative,
Main Office
Orange Grove Farm, Main Korung Rd
Banigala, Islamabad
Pakistan

AKR Foundation
Suite C-3 & C-4,
House #06,
Road # 109,
Gulshan-2, Dhaka-1212,
Bangladesh

Data will be transferred from Bangladesh and Pakistan to University of Edinburgh in the UK via a secure, password-protected online platform (DataSync) and then stored on a secure, password-protected DataVault / DataShare at University of Edinburgh for a minimum of 10 years following the end of the study. After 10 years, all paper records will be shredded and disposed of securely and electronic records will be permanently erased. All essential documents, including source documents, will be retained for a minimum period of 10 years after study completion.

The University of Edinburgh has a backup procedure approved by auditors for disaster recovery. There will be a separate archival of electronic data performed at the end of the trial, to safeguard the data, and in accordance with regulatory requirements.

27.2 QUALITY ASSURANCE AND CONTROL

27.2.1 Training

The following training procedures will be conducted to ensure quality control.

Person trained	Description
RA, Clinical TB care team	• CO/Cotinine testing • Consent procedure • Data collection and management • Safety and effectiveness assessments • Eligibility assessment
Country PI/ TM	Randomisation, Database management (data entry and querying procedures), management of logs/files/registers kept at the coordinating centre

27.2.2 CO and Cotinine testing

Data quality for CO and cotinine testing will be ensured through training, supervision and calibrating CO monitors. In this trial, the RA will perform cotinine dipstick tests on urine samples collected from the patients, which will be discarded after cotinine levels have been recorded on the CRFs.

27.2.3 Site Approval, Start-up Procedures and Ongoing Support

The following documentation must be in place and verified by the country PI's in order for a HC to be eligible to participate in the trial and to begin recruiting:

- A copy of signed Approval letter or Memorandum Of Understanding (MOU)
- A copy of the IRB approval (in line with local regulatory requirements)
- Completed Delegation of responsibilities Log and Contact Details (this could be incorporated in the MOU as per local procedures)

Once this documentation has been received, confirmation that the study may commence at that site will be sent to the site PI by the country PI.

Prior to commencing the study, each study site will be visited by the country TM or PI.

RA and the Clinical TB care team at the site will be trained in the data collection, data transfer, and filing and other trial procedures in order to comply with GCP. At this time the protocol will be reviewed in detail and the MOP provided.

Phone support, and visit support, will be provided to sites as per their requirements during the course of the study, mainly by the country TMs, with the support of the trials unit as appropriate.

27.3 DATA MONITORING

27.3.1 Central Data Monitoring at Edinburgh

Each country coordinating centre will be responsible for its own data entry and local trial management. Data will be entered into the trial database at the country coordinating centres. The sites/countries will retain the original CRFs. Data stored on the central database will be checked by personnel authorised by the trials unit for missing or unusual values (range checks) and checked for consistency within participants over time. If any such problems are identified, the country coordinating centres will be contacted and asked to verify or correct the entry. Changes will be made on the original CRF at the site/country and entered into the database at the country coordinating centre.

27.3.2 In-country Data Monitoring

The country coordinating centres will send reminders for any overdue and/or missing data with the regular inconsistency reports of errors to the RA at each participating site.

The country trial manager and/or PI will visit the trial sites to validate and monitor data. The frequency, type and intensity for routine monitoring and the requirements for triggered monitoring will be detailed in the trial specific operating procedures. The Quality Assurance plan will also detail the procedures for review and sign-off.

A detailed site initiation visit with training will be performed at each study site by the country PI or designated trial manager. The site initiation visits will include training in the study protocol, consent processes, and administration of the study interventions.

Monitoring and site training will be carried out at each site within intervals specified in the 'Quality Assurance Plan'. The monitoring will adhere to the principles of ICH GCP and will follow an agreed monitoring plan.

Delegated members of the research team will:

- confirm adherence to protocol
- review eligibility verification and consent procedures
- look for missed outcomes reporting
- verify completeness, consistency and accuracy of data being entered on CRFs
- evaluate medication accountability
- provide additional training as needed

The delegated members of the research team will require access to all (enrolled TB) patient medical records including, but not limited to, laboratory test results and prescriptions. The site PI (or delegated personnel) should work with the delegated members of the research team to ensure that any problems detected are resolved.

As part of the trial quality monitoring, the research assistants will make a weekly call to the participants (on provided number) to ask about SMS messages being received, and to remind about follow up visits.

27.4 DIRECT ACCESS TO PATIENT RECORDS

Site PIs should agree to allow trial-related monitoring by providing direct access to source data and documents as required. Such information will be treated as strictly confidential and will in no circumstances be made publicly available data controller along with any other entities involved in delivering the study that may be a data controller in accordance with applicable laws (e.g. the research site or partnering institution in the local country where the study is being conducted)

27.4.1 Data Breaches

Any data breaches will be reported to the University of Edinburgh Data Protection Officer who will onward report to the relevant authority according to the appropriate timelines if required. These Data Protection Officers in Bangladesh and Pakistan will also be notified of data breaches:

<u>Bangladesh</u>
Ms Samina Huque
Database Manager
ARK Foundation
Bangladesh
Email: huque1983@gmail.com

<u>Pakistan</u>
Mr Saeed Ansari
Database Manager
The Initiative
Pakistan
Saaed.ansaari@gmail.com

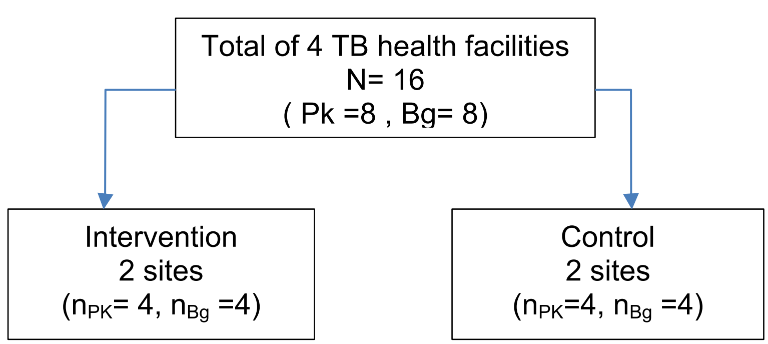
28 STATISTICS AND DATA ANALYSIS

28.1 SAMPLE SIZE CALCULATION

The study requires a total of 2,716 smokers with TB (~50 recruits from 48 health clinics [the clusters]) assuming that 10% do not provide primary outcome data. This includes the first 16 participants for the pilot study.

Pilot study:

The pilot study will be conducted in 4 randomly selected TB health facilities (2 sites in Bangladesh and 2 sites in Pakistan which will be assigned randomly to control and intervention) and in each site a total of n= 16 smokers with TB will be selected randomly.



Phase 3

The required sample size for Phase 3 (comparing A with C) considering 90% power, 5% significance level and assuming proportion achieving abstinence in usual care was 8% and using the mTB-Tobacco was 18% abstinence at 6 months and also considering the ratio of sample size (Intervention/Control = 2) was calculated as n=587.

The health facilities (clusters) would be randomised by ratio of 2:1 to mTB-Tobacco (intervention A), and standard care (Control), therefore a total of 27 site (TB health facilities) will be included in this phase which 18 clusters will be randomised to A and 9 clusters will be randomised to C in both countries.

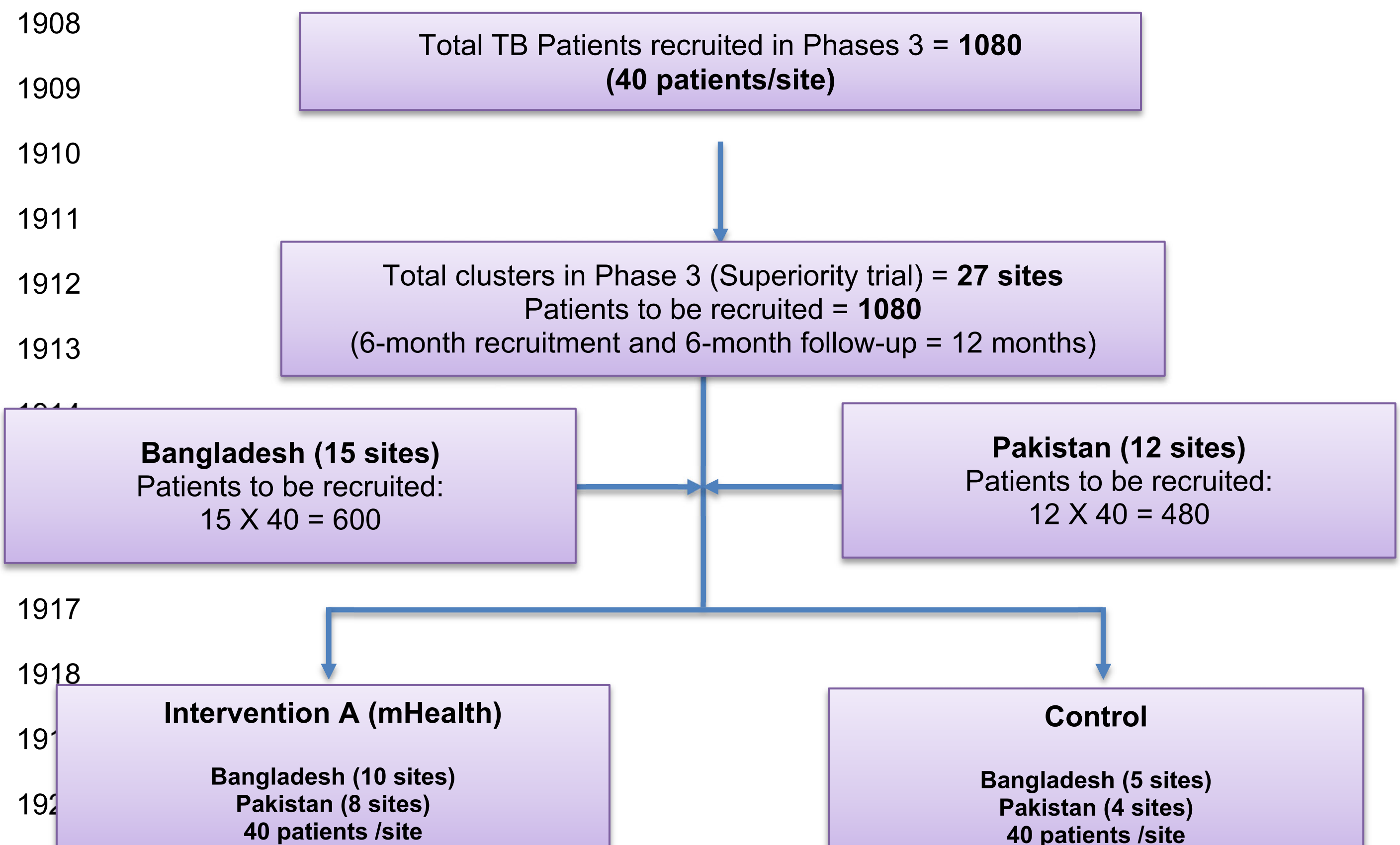
Considering 20% attrition rate:

- Final sample size: N= N0 + N0 *20%
- N= 587 + (587 * 0.2) = 704
- Total cluster (site) in Phase 3 = 27
- Sample per cluster = 704 /27 = 26

Since this study will be conducted as a cluster randomised trial (CRCT) and also considering 27 cluster and 26 participants per each cluster and ICC= 0.02 , the design effect was calculated as: DE = 1 + ρ (m-1), Where:

- m = number of subjects in a cluster and
- ρ = intra cluster correlation coefficient

DE = 1 + 0.02 (26-1) = 1.50
ESS = effective sample size= 704 * 1.50 ≈ 1080 (40 subjects for each site)



Phase 4

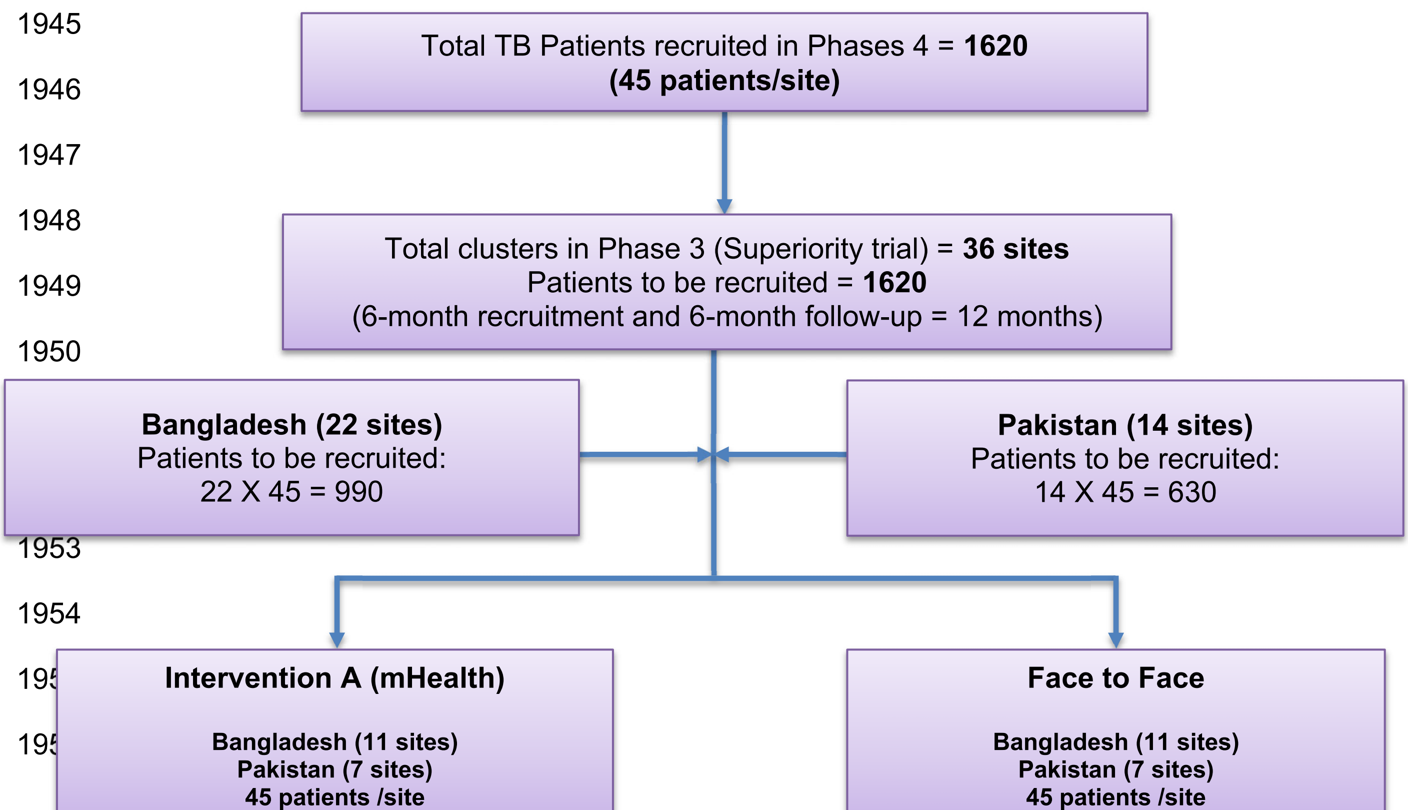
In phase 4 (If A was superior to C), considering 90% power, 1-sided 2.5% level to establish non-inferiority of mobile to face-to-face, comparing 18 clusters in each of A and B assuming face-to-face achieved 18% abstinence at six months and the non-inferiority margin was 8%. Assuming that just 2% would give up with no intervention at all, the established treatment effect of face-to-face over natural cessation over 6 months would then be 16% (18%-2%). The non-inferiority margin of 8% equates to the mobile intervention preserving at least 50% (8/16) of the established treatment effect of the face-to-face intervention (We will check above sample size assumptions mid-way through the recruitment period and adjust the sample size accordingly).

Considering the ratio of sample size (mobile intervention / face-to-face I =1) the final sample size was calculated as n=864.

Considering 20% attrition rate:

- Final sample size: N= N0 + N0 *20%
- N= 864 + (864 * 0.2) = 1036
- Total cluster (site) in Phase 4 = 36 (18 each arm)
- Sample per cluster = 1036 /36 = 29

DE = 1 + 0.02 (29-1) = 1.56
ESS = effective sample size= 1036 * 1.56 = 1620 (45 subjects for each site)
.



Phase 3 comparing A with C, would have 90% power at 5% significance level to compare 16 clusters randomised to A with 8 clusters randomised to C, assuming proportion achieving abstinence in usual care was 8% and using the mTB-Tobacco was 18% abstinence at 6 months. If A was superior to C, Phase 4 would then continue, and would have 90% power at a 1-sided 2.5% level to establish non-inferiority of mobile to face-to-face, comparing 20 clusters in each of A and B assuming face-to-face achieved 18% abstinence at six months and the non-inferiority margin was 8%. Assuming that just 2% would give up with no intervention at all, the established treatment effect of face-to-face over natural cessation over 6 months would then be 16% (18%-2%). The non-inferiority margin of 8% equates to the mobile intervention preserving at least 50% (8/16) of the established treatment effect of face-to-face. We will check above sample size assumptions mid-way through the recruitment period and adjust the sample size accordingly.

28.2 PROPOSED ANALYSES

28.2.1 Cost-effectiveness analysis

A health economic evaluation is included alongside the two phases to establish the value for money afforded by the mTB-Tobacco intervention. The cost of providing the interventions is recorded and calculated using local cost data. These costs include staff, premises and pharmaceutical products. We also record the wider use of public and private sector health care used by patients with a self-report service use questionnaire. A cost profile is calculated by applying country specific unit cost tariffs to quantities. Total intervention cost, costs of wider health care use and also patients' out of pocket expenditure are derived and a total cost applied to each patient in each trial arm.

In Phase 3 we will conduct an incremental cost-effectiveness analysis of mTB-Tobacco over and above usual care. Health Related Quality of Life (HRQoL) data will be collected using EQ-5D-5L to calculate Quality Adjusted Life Years (QALYs). Incremental cost effectiveness ratios will be presented in terms of i) cost per QALY and ii) cost per additional quitter as based on the secondary outcome of the trial. A 12-month time horizon is selected for the CEA in order to utilise the longer time window of health care use and changes in HRQoL. A full sensitivity analysis will be presented to assess the robustness of the results and the probability that mTB-Tobacco is cost-effective at a range of threshold values.

In Phase 4 (non-inferiority trial) we will record resource quantities and attribute unit costs to the mTB-Tobacco intervention and face-to-face behaviour support, inclusive of wider health system costs following the methodology of Phase 3. Costs for each arm are calculated and compared to a pre-defined acceptable cost difference. We will also carry out a cost-effectiveness analysis in Phase 4. Difference in costs and QALYs between treatment arms will be estimated to calculate the ICER based on cost per QALY and cost per additional quitter.

Underlying uncertainty around the decision to adopt the intervention will be assessed using non-parametric bootstrap re-sampling technique. Bootstrapping is an efficient method for calculating the confidence limits for the ICER as its validity does not depend on any specific form of underlying distribution. We will perform the bootstrap 5,000 replications and construct the 95% confidence intervals for the ICERs based on the bootstrapping results. Cost-effectiveness acceptability curves (CEACs) will be constructed based on the bootstrap iterations to estimate the probability that the intervention is cost-effective at different threshold values for one QALY.

The ICER, calculated in terms of the cost per QALY, will be used to assess the value of money afforded by the intervention over and above the control and draw conclusions with respect to the potential cost-effectiveness of the intervention.

To assess the impact of imputing missing data, we will also conduct a complete case analysis based on the participants who have both complete costs and QALYs at all timepoints, following the same analysis method as the primary analysis above. We will also conduct sensitivity analyses using pattern mixture modelling to examine the assumptions for multiple imputation methods.

28.2.2 Statistical analysis

Outcomes will be primarily analysed in a missing-at-random architecture by fitting generalised linear mixed models using a log link and assuming an underlying Poisson distribution with clusters treated as a random effect. All analyses will account for clustering through the random intercept model. The crude and adjusted relative risks (RRs) and corresponding 95% CIs will be estimated.

15. SAMPLE HANDLING, PROCESSING AND ANALYSIS

Urine sample will be collected from patients on their follow up visit who are using beside smoking other forms of tobacco (smokeless etc). The participants will be provided with containers and private premises to provide the sample. Nicotine dipstick test will be carried out by the research assistants on the sample and result recorded. The sample will then be discarded, hence no storage or shipped from research site to 3rd party/other institution/a UK based institution/ company is required. The sample analysis is not critical to the conduct of the trial and neither related to tertiary/exploratory endpoint data and objectives.

16. TRANSLATION

13.1 Translation Services

All trial documents (questionnaires, consent forms, participant information sheets, flipbook, mhealth messages) will be translated in-house. The participant-facing documents and SMS messages will be translated into local languages (Bengali for Bangladesh site and Urdu for Pakistan site) by the country-specific central research team. Then, another two research team members (one for each study site) will translate the local language versions back into English. Another two research team members (one for teach study site) will compare the back-translated versions with the original version for any discrepancies, and will resolve them before finalising the translated local versions. We will check the local language versions with the PPI group before they are submitted to the local ethics committees for approval.

17. ADVERSE EVENTS

In the context of this study, we think it is very unlikely that any Serious Adverse Event (SAE) would be related to the intervention or research procedures. The interventions are behavioural interventions and are not harmful. CO measurement and the quantitative data collection methods proposed present very minimal risks, if any risk, to participants. Therefore, no adverse events will be reported to the sponsor for this study

29 OVERSIGHT ARRANGEMENTS

29.1 INSPECTION OF RECORDS

Investigators and institutions involved in the study will permit trial related monitoring on behalf of the sponsor, ethics committees review, and regulatory inspection(s). In the event of monitoring, the Investigator agrees to allow the representatives of the sponsor direct access to all study records and source documentation. In the event of regulatory inspection, the Investigator agrees to allow inspectors direct access to all study records and source documentation.

29.2 STUDY MONITORING AND AUDIT

The Sponsor will assess the study to determine if an independent risk assessment is required. If required, the independent risk assessment will be carried out by the ACCORD Quality Assurance Group to determine if an audit should be performed before/during/after the study and, if so, at what frequency.

Risk assessment, if required, will determine if audit by the ACCORD QA group is required. Should audit be required, details will be captured in an audit plan. Audit of Principal Investigator research sites, study management activities and study collaborative units, facilities and 3rd parties may be performed.

29.3 STEERING COMMITTEE/DATA MONITORING COMMITTEE

Due to the low-risk nature of this trial an independent steering and monitoring committee will be set up to undertake the roles traditionally undertaken by the Trial Steering Committee (TSC) and Data Monitoring and Ethics Committee (DMEC). This committee will comprise of independent members including a Chair and at least two other independent members including a statistician. The independent members of the committee will be allowed to see unblinded data, but this data would not be reported to the other members of the research team. The role of this committee will include the reviewing of all serious adverse events which are thought to be treatment related and unexpected. The committee will meet at least annually or more often as appropriate.

13.4 STUDY COLLABORATORS/PARTNERS

- Harry Campbell, Professor of Genetic Epidemiology and Public Health, is the Chief Investigator for RESPIRE. He is responsible for the overall conduct of RESPIRE projects.
- Kamran Siddiqi, Professor in Global Public Health at the University of York, is the UK PI for this study and the overall lead researcher for this study. He is responsible for ensuring that the trial is conducted in accordance with the approved protocol.
- John Norrie, Professor of Medical Statistics and Trial Methodology at the University of Edinburgh, is the statistician for this study. He is involved in the design of the study and will make sure that data collected is analysed using appropriate methods and is interpreted correctly.
- Steve Parrott, Reader in Health Economics at the University of York, is involved in the design of the health economic component and will conduct the cost analysis of the data collected.
- Rumana Huque, Bangladesh trial site PI, is responsible for the conduct of the trial at the Banglaesh site. She will provide training to site staff on the delivery of the protocol, maintain oversight of the Bangladesh trial site and take appropriate action to ensure good recruitment, and the quality and timeline of the data collection.
- Amina Khan, Pakistan trial site PI, is responsible for the conduct of the trial at this site. She will provide training to site staff on the delivery of the protocol, maintain oversight of the Bangladesh trial site and take appropriate action to ensure good recruitment, and the quality and timeline of the data collection. She will be responsible for retaining clear, accurate and complete records of all trial documentation. She will report on all safety issues and suspected breaches.
- Ai Keow Lim, RESPIRE Programmes and Trial Manager, is the study trial manager and is responsible for the management and organisation of the trial. She will prepare sponsor and ethics committee submissions and ensure that appropriate approvals are obtained prior to the commencement of the study. She will work with the site PIs and coordinators to maintain oversight of the trial sites. She will submit proposed amendments to the protocol to the Sponsor and ethics committees.
- The Research Fellows/Trial Coordinators (to be recruited) will be responsible for identifying and recruiting potential research participants, supervise, monitor and coordinate the data collection, help maintain synergy between technical and support services at field sites, monitor field activities and report to site PI on identified issues solutions, supervise, coordinate and support field staff and organise field level meetings.
- Ms Samina Huque, Database Manager at ARK Foundation, is responsible for developing the data management plan and the management of the data collected. They will coordinate the design, build, test and validation of the case report forms and database. They will ensure timely returns of the data, entering the data onto the trial database and checking for inconsistencies and incomplete data. They will perform regular checks to validate and verify information on the database. They will work with RESPIRE Open Science, Data and Methodologies Platform Coordinator to ensure that data are transferred to UoE in safe and secure ways.
- Mr Saeed Ansari, Database Manager at The Initiative, is responsible for developing the data management plan and the management of the data collected. They will

coordinate the design, build, test and validation of the case report forms and database. They will ensure timely returns of the data, entering the data onto the trial database and checking for inconsistencies and incomplete data. They will perform regular checks to validate and verify information on the database. They will work with RESPIRE Open Science, Data and Methodologies Platform Coordinator to ensure that data are transferred to UoE in safe and secure ways.

30 GOOD CLINICAL PRACTICE

30.1 ETHICAL CONDUCT

The study will be conducted in accordance with the principles of the International Conference on Harmonisation Tripartite Guideline for Good Clinical Practice (ICH GCP). In addition to the principles of the ethics committee(s)/Institutional Review Boards (IRBs) who have reviewed and approved this study.

Before the study can commence, all required approvals will be obtained, and any conditions of approvals will be met.

This study will receive approval from the University of Edinburgh's Medical School Ethics Committee (EMREC) in addition to the Bangladesh Medical Research Council and the National Bioethics Committee-PMRC in Pakistan.

30.2 PRINCIPAL INVESTIGATOR RESPONSIBILITIES

The Principal Investigator is responsible for the overall conduct of the study at the site and compliance with the protocol and any protocol amendments. In accordance with the principles of ICH GCP, the following areas listed in this section are also the responsibility of the Principal Investigator. Responsibilities may be delegated to an appropriate member of study site staff.

30.2.1 Informed Consent

The Principal Investigator is responsible for ensuring informed consent is obtained before any protocol specific procedures are carried out. The decision of a participant to participate in clinical research is voluntary and should be based on a clear understanding of what is involved.

Participants must receive adequate oral and written information – appropriate Participant Information and Informed Consent Forms will be provided. The oral explanation to the participant will be performed by the Principal Investigator or qualified delegated person and must cover all the elements specified in the Participant Information Sheet and Consent Form.

The participant must be given every opportunity to clarify any points they do not understand and, if necessary, ask for more information. The participant must be given sufficient time to consider the information provided. It should be emphasised that the participant may withdraw their consent to participate at any time without loss of benefits to which they otherwise would be entitled.

The participant will be informed and agree to their medical records being inspected by regulatory authorities and representatives of the sponsor.

The Principal Investigator or delegated member of the trial team and the participant will sign and date the Informed Consent Form(s) to confirm that consent has been obtained. The participant will receive a copy of this document and a copy filed in the Principal Investigator Site File (ISF) and participant’s medical notes (if applicable).

30.2.2 Study Site Staff

The Principal Investigator must be familiar with the protocol and the study requirements. It is the Principal Investigator’s responsibility to ensure that all staff assisting with the study are adequately informed about the protocol and their trial related duties.

30.2.3 Data Recording

The Principal Investigator is responsible for the quality of the data recorded in the Case Report Form (CRF) at each Principal Investigator Site.

30.2.4 Principal Investigator Documentation

The Principal Investigator will ensure that the required documentation is available in local Investigator Site files ISFs.

30.2.5 GCP Training

All researchers will be encouraged to undertake GCP training in order to understand the principles of GCP. For participating in this trial, this will be a mandatory requirement for all researchers. GCP training status for all investigators should be indicated in their respective CVs.

30.3 DATA PROTECTION TRAINING

All University of Edinburgh employed researchers, students and study staff will complete the Data Protection Training through Learn.

30.4 INFORMATION SECURITY TRAINING

All University of Edinburgh employed researchers, students and study staff will complete the Information Security Essentials modules through Learn and will have read the minimum and required reading setting out ground rules to be complied with.

30.4.1 Confidentiality

All laboratory specimens, evaluation forms, reports, and other records must be identified in a manner designed to maintain participant confidentiality. All records must be kept in a secure storage area with limited access. Clinical information will not be released without the written permission of the participant. The Principal Investigator and study site staff involved with this study may not disclose or use for any purpose other than performance of the study, any data, record, or other unpublished information, which is confidential or identifiable, and has been disclosed to those individuals for the purpose of the study. Prior written agreement from the sponsor or its designee must be obtained for the disclosure of any said confidential information to other parties.

30.4.2 Data Protection

All Principal Investigators and study site staff involved with this study must comply with the requirements of the appropriate data protection legislation (including the European Union General Data Protection Regulation, the Data Protection Act 2018 in the UK and any relevant Data Protection laws in Bangladesh and Pakistan, respectively with regard to the collection, storage, processing, and disclosure of personal information.

Computers used to collate the data will have limited access measures via usernames and passwords.

Published results will not contain any personal data and be of a form where individuals are not identified, and re-identification is not likely to take place.

31 STUDY CONDUCT RESPONSIBILITIES

31.1 PROTOCOL AMENDMENTS

Any changes in research activity, except those necessary to remove an apparent, immediate hazard to the participant in the case of an urgent safety measure, must be reviewed and approved by the Chief Investigator.

Amendments will be submitted to the sponsor for review and authorisation before being submitted in writing to the appropriate ethics committees/IRBs, for approval prior to participants being enrolled into an amended protocol.

31.2 MANAGEMENT OF PROTOCOL NON-COMPLIANCE

Prospective protocol deviations, i.e., protocol waivers, will not be approved by the sponsor and therefore will not be implemented, except where necessary to eliminate an immediate hazard to study participants. If this necessitates a subsequent protocol amendment, this should be submitted to the appropriate ethics committees/IRBs for review and approval if appropriate.

Protocol deviations will be recorded in a protocol deviation log and logs will be submitted to the sponsor every 3 months. Each protocol violation will be reported to the sponsor within 3 days of becoming aware of the violation. All protocol deviation logs, and violation forms should be emailed to the sponsor at: QA@accord.scot

Deviations and violations are non-compliance events discovered after the event has occurred. Deviation logs will be maintained for each site in multi-centre studies. An alternative frequency of deviation log submission to the sponsors may be agreed in writing with the sponsors.

31.3 SERIOUS BREACH REQUIREMENTS

A serious breach is a breach which is likely to effect to a significant degree:

(a) the safety or physical or mental integrity of the participants of the trial; or

(b) the scientific value of the trial.

If a potential serious breach is identified by the Chief investigator, Principal Investigator or delegates, the sponsor (seriousbreach@accord.scot) must be notified within 24 hours. It is the responsibility of the sponsor to assess the impact of the breach on the scientific value of the trial, to determine whether the incident constitutes a serious breach and report to research ethics committees as necessary.

31.4 STUDY RECORD RETENTION

All study documentation will be kept for a minimum of 3 years from the protocol defined end of study point. When the minimum retention period has elapsed, study documentation will not be destroyed without permission from the sponsor.

31.5 END OF STUDY

The end of study is defined as the last participant's last visit.

The Investigators or the sponsor has the right at any time to terminate the study for clinical or administrative reasons.

The end of the study will be reported to the appropriate ethics committees/IRBs and sponsor within 90 days, (or within the specific timelines specified by the individual ethics committees/IRBs). If the study is terminated prematurely, the end of study will be reported to the relevant ethics committees/IRBs and sponsor within 15 days. The Investigators will inform participants of the premature study closure and ensure that the appropriate follow up is arranged for all participants involved.

End of study notification will be reported to the sponsor via email to researchgovernance@ed.ac.uk and to the relevant ethics committees who approved the study.

A summary report of the study will be provided to the appropriate ethics committee(s)/IRBs within 1 year of the end of the study (or within the specific timelines specified by the individual ethics committees/IRBs).

31.6 CONTINUATION OF TREATMENT FOLLOWING THE END OF STUDY

There are no plans to continue to provide the interventions (mTB-Tobacco and face-2-face behavioural counselling) beyond the trial. However, once policy makers have seen the trial results and have agreed to support one or both interventions, the investigators will support the respective TB programmes in implementing these interventions with the help of seeking implementation grants e.g., through Global Fund and TB Reach.

31.7 INSURANCE AND INDEMNITY

The sponsor is responsible for ensuring proper provision has been made for insurance or indemnity to cover their liability and the liability of the Chief Investigator and staff.

The following arrangements are in place to fulfil the sponsor responsibilities:

- The Protocol has been designed by the Chief Investigator and researchers employed by the University and collaborators. The University has insurance in place (which includes no-fault compensation) for negligent harm caused by poor protocol design by the Chief Investigator and researchers employed by the University.
- Sites participating in the study will be liable for clinical negligence and other negligent harm to individuals taking part in the study and covered by the duty of care owed to them by the sites concerned. The sponsor requires individual sites participating in the study to arrange for their own insurance or indemnity in respect of these liabilities.
- Sites in Bangladesh and Pakistan will be responsible for arranging their own indemnity or insurance for their participation in the study, as well as for compliance with local law applicable to their participation in the study.

32 STAKEHOLDER ENGAGEMENT

In Pakistan, we will engage with the TB programmes at both national and provincial levels and the tobacco control cell, including national and provincial focal points, from the start. In Bangladesh, we will engage with the National TB programme and the Directorate responsible for non-communicable diseases (including tobacco control) within the Ministry of Health. We will hold regular meetings with the respective focal persons in the afore-mentioned directorates and programmes and invite them to attend our steering committee meetings and dissemination events. The proposed study will strengthen our existing engagement with the respective TB control programmes in the two countries and help them in adopting the most cost-effective approaches to smoking cessation.

33 REPORTING, PUBLICATIONS AND NOTIFICATION OF RESULTS

33.1 CONFIDENTIALITY

The Parties each agree to use reasonable endeavours to keep confidential and not to publish or disclose in any way other than to those of its employees, students, directors, officers, advisors or representatives who have a need to know such information for the purposes of the Project:

- any Background IP of another Party identified as confidential at the time of disclosure; or
- any Results of another Party; or
- Joint Results;

(together the “**Confidential Information**”)

without the consent of the Party owning or controlling such Confidential Information for a period of 5 years from the conclusion of the Project.

33.2 AUTHORSHIP POLICY

Any publication or other dissemination of the Results (or any part of them) by any of the Parties shall not occur until Edinburgh has published the Results of the Project in the primary publication (the “Primary Publication”). Authorship of the Primary Publication shall be in accordance with normal academic practice. Each Party shall be entitled to publish articles directly arising from its solely owned Results. Prior to the publication of articles directly arising from the work of more than one Party on the Project, each Party shall endeavour to circulate proposed publications at least 30 days in advance of submission for publication. All publications shall acknowledge the funding made available for the Project by the NIHR as required by the NIHR Award Terms. Each Party retains the right to request (such request not to be unreasonably refused) the delay of a publication in order to seek Intellectual Property protection for Results generated in the course of the Project if publication would reasonably prejudice such protection. Such delay shall not exceed 3 months, unless mutually agreed between the relevant Parties. Notification of the requirement for delay in submission for publication must be received by the publishing Party within 30 days after the receipt of the material by the other Party/Parties, failing which the publishing Party shall be free to assume that the other Party/Parties has no objection to the proposed publication. These provisions clause shall survive termination or expiry of this Agreement for the period of 1 year.

No Party shall use the name or any trademark or logo of any other Party or the name of any of its staff or students in any press release or product advertising, or for any other commercial purpose, without the prior written consent of the Party(s).

Based on the study results, we will publish a range of high-quality materials and scientific papers from this trial. Therefore, it is hoped that all members of the team will engage with this aspect of the work, either by leading publications or through contributing to the writing process.

Opportunities to lead or participate in writing will be open to all members of the team who have made substantial contributions to the trial, whether through conceiving the work, securing funding, managing the project, conducting field work, generating, or interpreting data, or writing up findings.

Our publication process will be open and fair. This means that it will be transparent, so that all who wish to engage and/or contribute are aware of any opportunities or planned publications. However, authorship carries responsibilities, and all participants need to be aware of these. As per standard academic practice, authorship can only be rightly claimed if the ICMJE criteria are met. Where a manuscript or conference abstract is based on the methods and data of the trial, it may be appropriate to limit authorship to the members of the study team. However, after listing authors’ names, a suffix will be added, with the words ‘the RESPIRE Collaboration’. If the journal allows, all individually named authors who are members of RESPIRE should list “NIHR Global Health Research Unit on Respiratory Health (RESPIRE), The Usher Institute, University of Edinburgh, Edinburgh, UK” as one of their affiliations.

In order to promote inter-institutional collaboration within the trial, all manuscripts and conference abstracts will include authors from more than one institution, wherever possible. It is not expected that any one person within the team takes responsibility for policing adherence to the above authorship criteria. All members are expected to adhere to their own professional and/or academic codes of conduct, in line with the criteria above.

It is also important that the team members do not breach other ethical standards with respect to academic publishing. These include declaration of conflicts of interest, protecting confidentiality/anonymity, rules on gift authorship and ghost authorship, salami (segmented) publication and duplicate publication. (See http://www.wame.org for further information).

34 REFERENCES

1 WHO | Global tuberculosis report 2018. 2019; published online March 3. https://www.who.int/tb/publications/global_report/en/ (accessed Aug 29, 2019).

2 Zellweger J-P, Cattamanchi A, Sotgiu G. Tobacco and tuberculosis: could we improve tuberculosis outcomes by helping patients to stop smoking? European Respiratory Journal. 2015; **45**: 583–5.

3 Lin HH, Ezzati M, Murray M. Tobacco smoke, indoor air pollution and tuberculosis: a systematic review and meta-analysis. *PLoS Med* 2007; **4**: e20.

4 Lönnroth K, Castro KG, Chakaya JM, *et al.* Tuberculosis control and elimination 2010–50: cure, care, and social development. *Lancet* 2010; **375**: 1814–29.

5 Slama K, Chiang CY, Enarson DA, *et al.* Tobacco and tuberculosis: a qualitative systematic review and meta-analysis. *International Journal of Tuberculosis and Lung Disease* 2007; **11**: 1049–61.

6 Leung CC, Yew WW, Chan CK, *et al.* Smoking adversely affects treatment response, outcome and relapse in tuberculosis. *Eur Respir J* 2015; **45**: 738–45.

7 Yen YF, Yen MY, Lin YS, *et al.* Smoking increases risk of recurrence after successful anti-tuberculosis treatment: a population-based study. *Int J Tuberc Lung Dis* 2014; **18**: 492–8.

8 Marshall A-M, Barua D, Mitchell A, *et al.* Smoking prevalence among tuberculosis patients: A crosssectional study in Bangladesh and Pakistan. *Tob Induc Dis* 2020; **18**: 70.

9 Warsi S, Elsey H, Boeckmann M, *et al.* Using behaviour change theory to train health workers on tobacco cessation support for tuberculosis patients: a mixed-methods study in Bangladesh, Nepal and Pakistan. *BMC Health Serv Res* 2019; **19**: 71.

10 Siddiqi K, Khan A, Ahmad M, *et al.* Action to Stop Smoking In Suspected Tuberculosis (ASSIST) in Pakistan: a Cluster Randomized, Controlled Trial. *Ann Intern Med* 2013; **158**: 667–75.

11 Integrating Tobacco Control into Tuberculosis and HIV Responses. UNDP. https://www.undp.org/content/undp/en/home/librarypage/hiv-aids/integrating-tobacco-control-into-tuberculosis-and-hiv-responses.html (accessed Sept 26, 2019).

12 Dogar OF, Keding A, Gabe R, *et al.* Multicentre Randomised Double-Blind Placebo Controlled Trial of Cytisine for Smoking Cessation in Smokers with Tuberculosis. *The Lancet Global Health* 2020; published online June 24. http://eprints.whiterose.ac.uk/162580/ (accessed Aug 2, 2020).

13 Siddiqi K, Keding A, Marshall A-M, *et al.* Effect of quitting smoking on health outcomes during treatment for tuberculosis: secondary analysis of the TB & Tobacco Trial. *Thorax* 2021; published online July 16. DOI:10.1136/thoraxjnl-2020-215926.

14 Dogar O, Elsey H, Khanal S, Siddiqi K. Challenges of Integrating Tobacco Cessation Interventions in TB Programmes: Case Studies from Nepal and Pakistan. *J Smok Cessat* 2016; **11**: 108–15.

15 Organization WH, Others. Be he@ lthy, be mobile: a handbook on how to implement mTB-Tobacco. 2019. https://apps.who.int/iris/bitstream/handle/10665/325243/9789241515900-eng.pdf.

16 West R, Hajek P, Stead L, Stapleton J. Outcome criteria in smoking cessation trials: proposal for a common standard. Addiction. 2005; **100**: 299–303.

17 Dogar O, Keding A, Gabe R, Marshall AM, Huque R, Barua D, Fatima R, Khan A, Zahid R, Mansoor S, Kotz D, Boeckmann M, Elsey H, Kralikova E, Parrott S, Li J, Readshaw A, Sheikh A, Siddiqi K; TB and Tobacco Consortium. Cytisine for smoking cessation in patients with tuberculosis: a multicentre, randomised, double-blind, placebo-controlled phase 3 trial. Lancet Glob Health. 2020 Nov;8(11):e1408-e1417. doi: 10.1016/S2214-109X(20)30312-0. PMID: 33069301.

Changes between version v4 and v6

The key changes included adding ethics reference numbers, correcting definition of a secondary outcome (treatment failure), revising sample sizes, revising participant's inconvenience/travel allowance, adding a sub-study, including educational leaflet in the intervention and usual care group, and providing details on the weekly calls to participants. These are listed, as follows:

- The **reference numbers of the ethics committees** in Bangladesh and Pakistan were added in v6 - Page 0
- The **definition of treatment failure** – a secondary outcome- was corrected and "….who remained smear-positive at month 5…" was replaced by "…who remained smear positive at month 6…" in v6 – Page 6
- The **sample size** of the Phase 3 superiority trial (reported here) was increased from 888 participants in 16 intervention and 8 usual care clusters in v4 to 1080 participants in 18 intervention and 9 usual care clusters in v6. The sample size of the Phase 4 non-inferiority trial (not reported here) also went up from 1480 participants in 40 clusters to 1620 participants in 36 clusters in v6. The combined sample size of the two phases (superiority and non-inferiority trials) increased accordingly– page 7 in v6. The Diagram 1: Quit 4 TB Trial flow diagram on page 10 of v6 was also altered accordingly. In the sample size section 11.1 (from page 21-23 in v6), we updated the full sample size section for all phases.
- The participant's **inconvenience/travel allowance** was revised keeping in view the inflation and changes in transport cost in both the countries. This amount was paid as allowance at two study follow ups – Page 9 of v6.
- In v6, we introduced Technology Assessment Model as a **sub-study**. We added "At the end of Phase 3 Superiority Trial, 20% of the participants in the mTB-Tobacco intervention group in each country will be randomly selected and asked to complete a short Technology Acceptable Model (TAM) questionnaire. In Phase 4 Non-inferiority Trial, all participants in the mTB-Tobacco intervention group will be asked to complete the TAM questionnaire" – Page 13 of v6. The assessment table on page 13 was also changed accordingly. On page 14 and 15 of v6, we provided further details of TAM questionnaire, how this will be used with additional justification.
- On page 12 of v6, we added details on the **education leaflet** provided in both groups. We added "For phase 3, in addition to mHealth smoking cessation package, this group will also receive education leaflets from TB health professionals."
- On page 20 of v6, we added details on the **weekly calls** to the participants as follows, "As part of the trial quality monitoring, the research assistants will make a weekly call to the participants (on provided number) to ask about SMS messages being received, and to remind about follow up visits."

Statistical Analysis Plan

1 Introduction

The Statistical Analysis Plan (SAP) establishes the protocols for reporting trial progress and assessing effectiveness outcomes in the Quit4TB trial. This Statistical Analysis Plan (SAP) excludes analysis concerning cost-effectiveness and process evaluation which are explained in other documents.

The next parts provide an overview summary of the Quit4TB trial. The trial protocol document version 6.0 dated 20.Mar.2025 contains complete details along with the published protocol paper by Zahid M et al. 2025[1]

2 Study design

Using an adaptive design, this study will be conducted as a multi-center, cluster randomized, controlled trial consisting of four phases.

Phase 1 – PPI

Involves consultation with Patients and Public Involvement (PPI) group about the study processes. PPI members will also review and provide feedback on the participant facing materials.

Phase 2 - Pilot study

Testing feasibility of the study intervention and database

Phase 3 (superiority trial)

A multicenter Cluster Randomized Controlled Trial (cRCT) spanning 12 months, with 6 months allocated for recruitment and 6 months for follow-up. The trial comprises two arms: mTB-Tobacco (Intervention A) and usual care (Control),

Phase 4 (non-inferiority trial),

which will last for another 12 months (6 months recruitment and 6 months follow ups), we will compare mTB-Tobacco (intervention A) with face-to-face behavioural support (intervention B).

3 Intervention

Phase 3 (superiority trial)

[1] (Zahid M, Rahman F, Chowdhury A, Rana SH, Ansaari S, Lim AK, Danaee M, Boeckmann M, Parrott S, Norrie J, Khan A. mHealth intervention (mTB-Tobacco) for smoking cessation in people with drug-sensitive pulmonary tuberculosis in Bangladesh and Pakistan: protocol for an adaptive design, cluster randomised controlled trial (Quit4TB). BMJ open. 2025 Feb 1;15(2):e089007.

The mTB-Tobacco programme (intervention A) is designed to send short message service (SMS) content with two distinct areas of intervention. The first tries to instill behaviour change in patients to quit the habit of tobacco use and the second includes a combination of supportive motivational and informative messages through the period of TB treatment. The overall outcome being that a person is able to both successfully quit tobacco consumption and complete TB treatment.

A generic intervention has already been developed by the investigators in collaboration with the WHO mHealth team, available here: https://www.who.int/publications/i/item/a-handbook-on-how-to-implement-mtb-tobacco. This resource provides guidance for national TB control programmes and organisations responsible for delivering TB control in adapting and implementing a bespoke mTB-Tobacco programme. The resource includes a set of SMS messages to inform TB patients about the hazards of tobacco use and encourage them to quit. Furthermore, these messages are also designed to enhance treatment adherence, increase healthy behaviours, and reduce potentially harmful behaviours in TB patients in order to improve their health outcomes.

To make the intervention ready for the trial, the research team will set up a reference group (TB and tobacco experts, TB programme managers and clinicians) to support, and inform the adaptation of mTB-Tobacco intervention. The group will first select a set of SMS messages (from the abovementioned resource) and draft their transmission schedule (the number of messages, their frequency and time intervals). Once a plan of the mTB-Tobacco programme has been developed, it will go through iterative stages of pre-testing, piloting, and refinement.

In phase 1, we will organise PPI forum meetings to ask them to review our patient facing materials including SMS messages for comprehension, language, and tone. The PPI group will comprise of six-eight members including TB patients, carers and lay members of public, both in Bangladesh and Pakistan. Messages that are considered as unclear or inappropriate for the target group would be rewritten.

Once the messages have been reviewed by the PPI group, the intervention will be tested (as part of the pilot study) with a total of 16 patients (8 each in Bangladesh and Pakistan). The pilot will be evaluated using two sets of data: users' self-reported experiences and their real-time engagement with the programme. Users’ experiences can be collected through face-to-face/phone-based interactions. Key questions would focus on users’ experience of taking part in the mTB-Tobacco programme; the clarity, quantity, timing and frequency of messages; what was good about the programme and what was not; completion or non-completion the programme; and any effect on their attitudes or target behaviour(s). Users’ real-time engagement will be evaluated using computer records from the programme used for launching mTB-Tobacco. The pilot study is embedded in the superiority trial (phase 3). The pilot participant data will be treated with the same processes and governance as the main trial participant data.

Once mTB-Tobacco is piloted, but before continuing to superiority trial, the programme will go through a refinement process (a study pause) that will address issues highlighted during the pilot and make adjustments to the intervention, if needed.

At the end of above exercise, the intervention will be ready for the superiority trial and will comprise of: (a) a set of SMS messages translated in Bangla and Urdu, respectively; (b) a schedule of transmission; (c) an automated digital SMS messaging platform ready to send these messages as per schedule; and (d) a dedicated team to supervise and monitor the transmission of these messages.

The SMS messaging will only store unique ID number for the trial. No other personal or identifiable information will be stored. Each participant in the intervention group will receive a total of 134 SMS messages over a period of 6 months. In the first month, the frequency of messages would be 3 to 4 messages per day, in next month the frequency would reduce to 1 message per day and in the last four months there will be 1 message sent per month.

We are using REDCap for managing the data for this trial. The mHealth app will be linked to REDCap. At the screening stage, participants who meet all inclusion criteria and do not meet any exclusion criteria will be eligible to take part in the study. They will be given the participant information sheet and consent form. If they consent to take part in the study, their demographic details (including active mobile number) will be recorded on the REDCap database. If the participant is marked as belonging to the intervention arm, then his/her contact details/mobile number will be captured from the database by the mobile SMS service to send mHealth messages as per the schedule. The participants belonging to the intervention arm will start receiving the SMS messages, whereas those belonging to control arm will not receive any SMS.

The mobile SMS service is provided by a third-party service. A contract of services for the development of the mTB-Tobacco program application has been signed between The Initiative and Biter Tech (a third-party). Private Limited. Section A General Terms & Conditions Point 7 of the sub-contract (highlighted in yellow) mentions that “The Consultant is not allowed to keep or sell any project related data. The Consultant agrees to keep confidentiality of project data, and, to cause its employees and agents to keep confidentiality of any data available to them. The consultant will not access or check participant’s personal data unless it is necessary in connection with the services provided under this Agreement, or otherwise consented to, in writing.”

Phase 4 (non-inferiority trial)

This consists of an adapted version of a pre-developed and proven effective intervention to help people to quit smoking and smokeless tobacco use.10 This consists of two face-to-face sessions delivered at day 0 and day 5 (+2) and which last 10 and 5 minutes respectively. The sessions will be structured using an educational flipbook; the session on day 0 will be aimed at encouraging tobacco users to see themselves as non-users and set a plan for a quit date five days later; with a session on the quit date (day 5) to review progress. Further encouragement and support (if needed) will be offered at a subsequent visit at week 5.

The TB health professionals will be provided with patient education leaflets to be given out to all patients allocated to the respective trial arm. Leaflets for patients and others in the house contain information on the harmful effects of tobacco and advice on stopping its use.

4 Objectives of the study

Primary Objective

To evaluate the effectiveness and cost-effectiveness of mTB-Tobacco in achieving continuous abstinence for a minimum of six months in individuals diagnosed with both tuberculosis (TB) and daily smoking habits.

Secondary Objectives

- To investigate the effectiveness and cost-effectiveness of mTB-Tobacco in enhancing adherence to TB treatment among individuals who smoke daily.

- To assess the impact of mTB-Tobacco on clinical outcomes in individuals with both TB and daily smoking habits, considering factors such as disease progression, severity, and overall treatment response.

5 Randomization

In this Cluster Randomized Controlled Trial (cRCT), interventions will be administered at the group level, specifically in health facilities.

- Phase 2 (pilot study)
- Phase 3 (superiority trial),
- Phase 4 (non-inferiority trial)

The randomization process involves the allocation of health facilities (clusters) in a 2:2:1 ratio to the mTB-Tobacco intervention (Intervention A), face-to-face intervention (Intervention B), or standard care (Control). To ensure an unbiased allocation sequence, an independent statistician, situated at the University of Edinburgh, will employ computer-generated random-number lists.

6 Blinding

In this study, we used a blinding methodology to guarantee that specific key individuals remained uninformed about the allocation of treatments. The allocation sequence will be determined by an independent statistician, who has no knowledge of the participating facilities. Nevertheless, as the SMS receipt is a noticeable component, participants will be able to determine whether they are receiving the SMS or not. Therefore, it is not possible to achieve total blindness at the participant level in this circumstance. In addition, the principal investigator (PI) and the research teams are aware of the specific cluster assigned to each treatment group. However, steps will be taken to potentially prevent the data analyst from knowing the treatment assignments in order to reduce bias during the analysis of the data. Thus, the blinding in this trial will occur during randomising the clusters to intervention or control by an independent statistician and also at the level of data analyst while analysing the data.

7 Study Population

The trial targets populations in Bangladesh and Pakistan, countries with elevated burdens of tuberculosis (TB) and tobacco-related diseases. In Bangladesh, there are 357,000 TB cases with 47,000 deaths annually and a tobacco prevalence of 35%. In Pakistan, there are 562,000 TB cases with 44,000 deaths, and a tobacco prevalence of 20%. The trial focuses on recruiting TB health facilities (clusters) in the public sector, specifically from the Dhaka division in Bangladesh and the Punjab province in Pakistan. This strategic selection aims to address the significant health challenges posed by TB and tobacco use, ensuring a diverse and representative study population.

Country	TB (per 100,000)	TB cases (n)	TB death s (n)	Tobacco prevalence
Bangladesh	221	357,000	47,000	35%
Pakistan	265	562,000	44,000	20%

8 Sample Size Calculation

The study requires a total of 2700 smokers newly diagnosed with pulmonary TB (~43 recruits from 63 TB centres, the clusters) assuming that 20% might not provide primary outcome data. This sample also includes the first 16 participants taking part in the pilot study, in case the intervention needed major refinement then we would be unable to use their data.

The required sample size for the superiority trial (phase 3) was determined by using a power of 90%, a significance level of 5% and expected abstinence rates of 18% for mTB- Tobacco and 8% for normal treatment after 6 months.13 The study will comprise a total of 27 TB centres, considering the ratio of intervention to control as 2:1. After considering a 20% attrition rate, it was determined that the sample size would be 704 participants. This would result in an average of around 26 individuals in each cluster. Given that the study will be carried out as a cRCT with an intracluster correlation coefficient of 0.02, the design effect was computed to be 1.50. This results in an effective sample size of around 1080 people (or 40 subjects per site).

In the non-inferiority trial (phase 4), considering 90% power, one-sided hypothesis and 2.5% significance level to establish non-inferiority of mTB-Tobacco intervention to face- to-face behavioural support, comparing 18 clusters in each treatment arm assuming face-to- face achieved 18% abstinence at 6 months and the non- inferiority margin was 8%. Assuming that just 2% would give up with no intervention at all, the established treatment effect of face-to- face behaviour support over natural cessation over 6 months would then be 16% (18–2%). Given that the study will be carried out as a cRCT with an intracluster correlation coefficient of 0.02, the design effect was computed to be 1.50. This results in an effective sample size of around 1620 people (or 45 subjects per site) in phase 4.The non-inferiority margin of 8% equates to the mobile intervention preserving at least 50% (8/16) of the established treatment effect of face-to-face.

9 Outcomes and data collection

10 Primary Outcome(s)

Abstinence

The primary endpoint will be biochemically verified continuous abstinence at 6 months post randomisation. Abstinence is defined as self-report of not having used more than 5 cigarettes, bidis, water pipe sessions since the quit date, verified biochemically by a breath carbon monoxide (CO) reading of less than 10 ppm at month 6.

In case of concomitant smokeless tobacco use, biochemical verification will be carried out using cotinine strips to detect cotinine (a nicotine metabolite) level in urine samples. Reading of the strip will indicate a cotinine level (0-6), where level <3 (i.e., equivalent to 100-200ng/mL cotinine) will be considered tobacco abstinence.

When a patient self-reports abstinence with an elevated CO level of >10ppm (and cotinine levels in concomitant users indicating active tobacco use), the biochemical verification will supersede the self-report and the patient will be defined as a tobacco user.

11 Secondary Outcome(s)

12 Adherence to TB Treatment:

All registered TB patients' medication logs (for anti-TB medication) are recorded on the 'Treatment Support Card'; a copy of this card would be requested from the TB paramedic by the RA and attached with the patient CRF.

13 TB Programme outcomes:

the proportion of treatment success (including cured and completed treatment), treatment failure, defaulted and died will be recorded from TB register (TB03) at month 6. The definitions of these outcomes are as follows:

- Cured: 'A patient who was initially smear-positive and who was smear negative in the last month of treatment (at month 6) and on at least one previous occasion'
- Completed treatment: 'A patient who completed treatment (at month 6) but did not meet the criteria for cure or failure.'

- Treatment failure: ‘A patient wo was initially smear-positive and who remained smear-positive at month 5 or later during treatment’
- Defaulted: ‘A patient whose treatment was interrupted for 2 consecutive months or more’
- Died: ‘A patient who died from any cause during treatment’
- Relapse: ‘A patient who was previously treated for TB, was declared cured or treatment completed at the end of his most recent course of treatment, and is now diagnosed with a recurrent episode of TB (either a true relapse or a new episode of TB caused by reinfection)’

14 TB Outcomes

- Clinical TB score
- Sputum conversion results
- Adherence to TB treatment
- TB treatment outcome (success, failure, default, relapse or death)

The researchers assessed signs and symptoms for TB score (supported by the clinical team), while routine reporting on TB cards/registers maintained by the TB programme provided the remaining TB outcomes.

15 Nicotine Dependency

- *Mood and Physical Symptoms Scale (MPSS)* (West and Hajek 2004): The MPSS scale assesses withdrawal symptoms. Participants rate how they feel on five domains, using a 5-point scale: depression, irritability, restlessness, hunger, and poor concentration. These items can be summated to give an overall score ranging from 5 to 35. High values on these scales are indicative of high withdrawal symptoms.
- *Strength of Urges To Smoke (SUTS)* (Fidler et al. 2011): The SUTS scale assesses the presence and strength of urges to smoke using two questions. The resulting score ranges from 0 (no urges) to 5 (extremely strong urges).

16 Adverse Events

Adverse events (AEs) were collected up to week 9 and self-reports recorded on the ‘AEs review’ checklist: Nausea, Diarrhea; Dry mouth; Epigastric pain; Headache; Insomnia; Abnormal dreams; Irritability and moodiness; Anxiety; Tachycardia or palpitations Musculoskeletal pain

17 Socio-demographic

Participant characteristics were collected at baseline, including: age, gender, marital status, highest level of education, occupation, form of tobacco use and quit history

The recruiting site and country will be recorded for each patient

18 EQ-5D-5L Quality of Life (Economic Outcomes)

The EQ-5D-5L was collected as a standardised measure of current health status developed by the EuroQol Group for clinical and economic appraisal. The EQ-5D consists of five questions, each assessing a different quality of life dimension (Mobility, Self-care, Usual activities, Pain/Discomfort and Anxiety/Depression). A weighted and population referenced summary index is derived and will be reported and analysed as part of the health economic analysis.

Table 1: Data Collecti

Assessment	Screening	Baseline	9 Week	6 Months
Assessment of eligibility criteria	x			
Written informed consent	x			
Socio demographic characteristics		x		
Tobacco use and quit history		x		
Self-reported tobacco use/abstinence status				x
CO test, Cotinine test				x
Mood and Physical Symptoms Scale (MPSS)		x		x
Nicotine dependency (SUTS, HTUI)		x		x
TB outcomes (Clinical TB Score, Symptoms, Sign)		x	x	x
Economic Evaluation		x		x
EQ-5D-5L		x		x
Adherence to TB Treatment			x	x
Sputum smear microscopy			x	x
Adverse events review			x	x
m-TB usage (SMS)			x	

on Schedule

19 Data management

20 Documentation

Contact information of participants and their parent/guardian (applicable if the age of participants is below legal age) such as name, phone number and addresses will be stored in hardcopies (logs and sheets). Other data (stated in the section above) including phone numbers will be collected using REDCap mobile app after getting their consent. eCRF designed in REDCap will carry additional instructions/hints for FRAs (Field Research Assistants) which are supporting narratives related to questionnaires. An eCRF may consist of different sections.

21 Structure of datasets:

Under this project, multiple datasets will be created. eCRFs of Baseline, Follow-up 1 (Week 9) and Follow-up 2 (Month 6) are separated by countries. eCRFs of Screening and Change in Status are combined (both countries are collecting data in a single eCRF).

For the uniformity of the dataset, appended[2] data will be created after cleaning. A country-wise version of the dataset of different eCRF may also be generated later for ease of usage.

In addition to this, a wide-format-master merged dataset containing data from different times/periods/rounds will be created at initial stage. For the recognition of variables from each round, prefixes will be used as variable identifiers.

For ease of multivariate analysis (for some selected variables), a long-format-master dataset from three time point datasets of baseline, week 9 and month 6 will be created at later stage of data analysis. For both wide and long format master data, a data matrix including labels, hints and other key information (e.g., skip logics, hints, relevance) will be prepared. This will work as a data navigation tool.

22 Data Correction and Cleaning:

Once the data is collected through TAB from each study site (i.e., Bangladesh and Pakistan) using the eCRF on the REDCap app, they will be sent and archived in the REDCap server. It can then be downloaded in CSV, SPSS, STATA format by the respective focal person of each study site. After that the collected data can be checked for consistency regularly (every day at the beginning 2-3 weeks of the data collection period and at least once every 2 to 3 days in later period) by looking into the excel file/ CSV file/ frequency distribution of the variables. If any inconsistency or a query is found and raised regarding a variable (e.g., typo mistake, missing, outliers) then that will be identified with the trial number of study participant(s). The respective focal person of the study site will keep record of the inconsistency or data query in the data query, change request and change made log (named ***Data Query, Change Request and Changes Made Form***). The source of a query should be clear from the query identifier, i.e., QBX for queries raised in Bangladesh, QPX for queries raised in Pakistan. Query numbers should be logged sequentially from 1 to 999, so each query has a unique reference which can be referred to in all communication. For the data query identified data management team members of respective study site will meet with the data collectors and will note down if any correction required for the identified errors.

To correct any errors, the study participant trial number, REDCap ID and types of variable(s) to be amended must be downloaded in the mobile or tab in the REDCap app before making any corrections. After doing the corrections the corrected record must then be saved and uploaded to the respective records in the server. The Bangladesh team will complete this task based on the Data Query, Change Request and Changes Made Form.

In case of any corrections done and saved in REDcap server database for trial number or patient’s ID, type of study sites (intervention or control) and contact number of patients then this will be communicated with the mhealth app managing vendor and respective focal person/trail coordinator at Pakistan study site (who have admin access of mhealth app).

23 Location of Data and Associated Files

All eCRF data will be stored in a secured server of REDCap managed by ARK Foundation. The data then will be downloaded in local computer which is password protected and then will be checked for consistency and cleaned if necessary. The back-up of raw and cleaned data of eCRF in CSV, SPSS, STATA format will be stored in the University of Edinburgh (UoE) DataSync (a

Dropbox-like file hosting service). The hard copies including consent forms, different logs and tracker sheets will be preserved in respective country offices. The project will also create different documents (e.g., protocol, manuals, research/site administrative logs, data codebooks, data matrix, data management plan, and query sheet) associated with research and data management. The data (for both soft and hard data) and document storage time span of this project will be 10 years.

While taking back-up in DataSync, a date mark will be added with folder name. Format of the date mark is: YYYYMMDD (reverse date format). Here, Y represents Year, M represents Month and D represents Day. E.g., Data backup of 6th July 2024. Folder name will be: quit4tb_backup_clean_20240706.

For sending mhealth intervention message there will be an mhealth app and related databases which will extract data of recruited patients integrating with REDcap server through a secured API.

24 Analysis

25 Interim Analyses

At the end of phase 3 superiority trial, an interim analysis will be done by an independent statistician to assess the effectiveness of the mTB-Tobacco intervention. The result will decide whether the trial needs to proceed to phase 4 non-inferiority trial or be terminated early.

26 Descriptive analysis

Descriptive statistics covering demographic information, clinical data, and tobacco use will summarize baseline characteristics of the participants. While means and standard deviations or medians and interquartile ranges will summarize continuous variables depending on their distribution, frequencies and percentages will characterize categorical variables. Exploratory data analysis (EDA) will be executed before formal analysis to investigate key variable distributions and to explore missing data patterns as well as detect potential outliers. To determine the normality of continuous variables researchers will visually inspect histograms and Q-Q plots and calculate measures of skewness and kurtosis. Following these steps will guide researchers in selecting the right statistical tests and modelling approaches for future analysis.

27 Trial Progression

The flow of participants from screening and randomisation to follow-up and analysis of the trial will be presented in a CONSORT flow diagram (see Figure 1).

28 Excluded Patients

Exclusion reasons will be descriptively summarized for all patients considered ineligible, with frequencies and percentages provided for each trial arm.

29 Baseline Data

Baseline characteristics will be presented descriptively in total and by trial arm for all randomised participants.

30 Primary outcome Analysis

The primary outcome will be biochemically verified continuous abstinence at 6 months post randomisation. Agreement between self-reported and biochemically verified abstinence will be reported by trial arm. The primary outcome (self-reported and verified continuous abstinence at 6 months) will be measured by reporting both the number and proportion of abstinent patients for each trial arm. Statistical analysis for complete case, will provide group differences by calculating both risk difference and relative risk figures together with their 95% confidence intervals. We will run a mixed-effects logistic regression model that treats site as a random effect to determine the treatment effect which we will express through relative risk with its corresponding 95% confidence interval and p-value.

The primary outcome analysis will utilize an intention-to-treat (ITT) method. Patients who drop out or have incomplete data will be regarded as continuing smokers in line with the Russell Standard methodology (West et al., 2005).

31 Secondary Outcome Analyses

32 TB Outcomes

TB outcomes of TB score, sputum conversion, treatment adherence and programme outcomes, will be analysed using appropriate descriptive and inferential statistical models for continuous or categorical data using all available time points.

33 Adverse Events

The chi-squared test will be applied to compare adverse events between the mHealth group and the control group. Frequencies and percentages were reported. The chi-squared test compared the number of patients with any adverse event and serious adverse events between groups when at least ten patients had one or more events. The results for low frequencies adverse events will be adjusted using exact test.

34 EQ-5D-5L Quality of Life

The EQ-5D-5L quality of life as a secondary outcome will be analyzed using total score (based on ……...). The total scores at two time points baseline and six months following randomisation) will be analyzed through a mixed-effects model that includes site clustering as a random effect to account for possible intraclass correlation. Fixed effects in the model will include treatment group and time periods as well as their interaction. The study will demonstrate treatment impacts on EQ-5D-5L scores through estimated marginal means and intergroup differences which include 95% confidence intervals and p-values.

35 Sensitivity Analyses

36 Adjusted Analyses

The analysis will employ a mixed-effects logistic regression model that factored in site as a random effect to handle clustering by site. The fully adjusted analysis models will be controlled for by adjusting for sociodemographic variables alongside tobacco use form and other baseline characteristics (such as education level and smoking duration).. The models will account for site as a random effect to properly adjust for clustering effects. For every model:

- The treatment effect will be expressed as a Relative Risk (RR) and 95% Confidence Intervals (CIs).
- Categorical variables will be treated as factors, and continuous variables were analyzed per unit increase.

37 Analysis excluding missing data

The primary logistic regression will be repeated for patients with available outcome data only, i.e. excluding patients for whom primary outcome data was missing. The number and proportion of abstinent patients will be reported by trial arm for self-reported and CO- verified abstinence. The treatment effect coefficient will be presented with 95% confidence interval and associated p-value.

38 Sub-group Analyses

We will perform sensitivity analyses to evaluate how stable the primary treatment effect remains across important subgroups. The study will segment the population for subgroup analysis based on age (<40 years vs. ≥40 years), education level (no formal education, primary education, more than primary education), occupation (active employment/business vs. dependents/retired), gender (male vs. female), and smoking duration (<24 years vs. ≥24 years). Each subgroup will have its own mixed-effects model applied while accounting for clustering at site level through a random intercept. We will compute both the relative risk (RR) and the 95% confidence intervals (CIs) for smoking cessation. Researchers will assess the treatment effect's consistency between subgroups through qualitative comparison of RRs and CIs. The researchers will disclose any missing confidence intervals while analyzing subgroup results with caution due to limited statistical power alongside their exploratory analysis nature. Sensitivity analyses enable researchers to evaluate if primary results maintain robustness when examining various sociodemographic and smoking-related characteristics.

39 Other Analyses

The research team or protocol changes can initiate further additional analyses. The exploration analyses planned will involve additional subgroup analyses along with sensitivity analyses and post hoc assessments based on emerging research questions. Documentation and separate reporting of these analyses will occur after justification distinct from the primary and secondary analyses that were originally specified.

40 Software

All statistical analyses will be done using the open-source R software (R Foundation for Statistical Computing) and JASP software (version 0.19.3; JASP Team, 2024).

41 Appendices

42 Planned Tables and Figures

The following selected tables and figures are intended as templates for the statistical analysis report. The final tables and figures may differ in some details from the templates proposed here.

Socio-demographic & Background

Variable	Level	m-Health	Control	χ^2	p value
Gender	Male				
	Female				
Education	No formal education				
	Primary				
	Middle				
	Secondary				
	Higher				
Marital status	Single				
	Married				
	Separated/Divorced /Widowed				
Occupation	Active in Job/Business				
	Dependents				
	Retired				
TB Stage	Stage 1				
	Stage 2				
	Stage 3				
Reading SMS	Yes				
	No				
Age	Mean (SD)				
BMI	Median(IQR)				

TOBACCO USE AND QUIT HISTORY

Types of smoking	Response	m-Health n(%)	Control n(%)	χ^2	P value
Cigarettes					
Bidi (hand rolled cigarette)					
Hookah (water-pipe)					
Electronic cigarette					
Smokeless tobacco products					
Other					

Tobacco Use

Variable	Types	Group	n	Median (IQR)	Mean (SD)	Z value [a]	P value
Tobacco Use (Days)	Cigarettes	m-Health					
		Control					

	Bidi	m-Health					
		Control					
	Hookah	m-Health					
		Control					
	Smokeless tobacco	m-Health					
		Control					
Daily Tobacco Use (average number)	Cigarettes	m-Health					
		Control					
	Bidi	m-Health					
		Control					
	Hookah	m-Health					
		Control					
	Smokeless tobacco	m-Health					
		Control					

Smoking Behavior and Cessation Attempts

Variable	Response	m-Health n(%)	Control n(%)	$\chi 2$/t/z	P value
Smoking inside home [a]					

Attempted to quit in the past [a]					
Smoking duration [c]					
Age starting [b]					

Strength of Urges To Smoke (SUTS)

Variable	Group	n	Median (IQR)	Mean (SD)	Z value [a]	P value
Urge to smoke in the past 24 hours**)**						
How strong have urges been?						

Primary outcome (Abstinence**)**

Mixed model

Outcome	**Trial Group**	**n/N(%)***	**Proportion (95% CI)**	**RR (95% CI)**	**Adjusted RR (95% CI)**	**ICC**
Self-reported 9 week	**m-Health**					
	Control					
Self-reported last 7 days	**m-Health**					
	Control					
Self-reported	**m-Health**					
	Control					
CO & Cotinine Test	**m-Health**					
	Control					
CO & Cotinine Test (ITT)	**m-Health**					
	Control					

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sub-groups for verified abstinence at 6 months

Variable	Category	RR	CI95%
Age	<40		
	>40		
Education	No formal education		
	Primary (classes 1-5)		

	More than Primary (classes >6)		
Occupation	Active in Job/Business		
	Dependents and retired		
Gender	Male		
	Female		
Smoking Duration	<24 Years		
	>24 Years		

43 Glossary

AE	Adverse Event
API	Application Programming Interface
cRCT	Cluster Randomized Controlled Trial
CI	Confidence Interval
CO	Carbon Monoxide
CONSORT	Consolidated Standards of Reporting Trials
CRF	Case Report Form
EDA	Exploratory Data Analysis
EQ-5D-5L	EuroQol 5-Dimensions 5-Levels Questionnaire
FRA	Field Research Assistant
ICC	Intraclass Correlation Coefficient
IQR	Interquartile Range
ITT	Intention to Treat
mHealth	Mobile Health
MPSS	Mood and Physical Symptoms Scale
PPI	Patient and Public Involvement
PI	Principal Investigator
ppm	Parts Per Million
RA	Research Assistant
REDCap	Research Electronic Data Capture
RR	Relative Risk
SAP	Statistical Analysis Plan
SMS	Short Message Service
SUTS	Strength of Urges to Smoke
TB	Tuberculosis
TTFC	Time to First Cigarette
UoE	University of Edinburgh

mTB-Tobacco Intervention SMS log

Day Received		**Message**	**Character Count**
Day Reruited Q (-8)[3] Tx (+1)[4]	Generic (for all)	1Welcome to mTB-Tobacco! Congratulations on deciding to QUIT tobacco use and improve your chances of successfully recovering from TB. Time: At time/day of enrolment	133
	Generic (for all)	2Well done for joining this programme. We will help you quit tobacco and adhere to TB medication by giving useful tips throughout the treatment course. Time: 2 mins after first message	151
	Generic (for all)	3You will get up to 5 text messages every day to help you successfully recover from TB. All messages received in mTB-Tobacco will be free. Time: 2 mins after third message	146
		mTB-Tobacco intervention phase!	
Q (-7) Tx (+2)	Generic (for all)	4TB germs spread in the air when TB patient coughs, sneezes or spits. TB is curable! You must take the medicine regularly for full [*xx*] months as your doctor tells. Time: 9:00 am in morning	163
	Generic (for all)	5Deciding to quit tobacco for good is a great step. You should feel proud of yourself. We'll send you tips to help you get ready! Time: 12:00 pm	128
	Generic (for all)	6The best way to quit tobacco use is to stop completely. To help you quit, we have set a date for you, 7-days from today, to stop using tobacco completely. Time: 5:00 pm	159
	Generic (for all)	7Remember, it will **NOT** be helpful if you: • slowly decrease tobacco use • use tobacco once in a while • or replace it with other kinds of tobacco such as [*waterpipe, bidi, smokeless tobacco*] Time: 7:00 pm	159
Q (-6) Tx (+3)	Generic (for all)	8Do you know that tobacco use can stop your recovery from TB? People who do not quit are more likely to have heart disease and cancer. Time: 9:00 am	139
	Generic (for all)	9Quitting tobacco will help with your recovery! The benefits of quitting will start within days and will continue during and after treatment. Time: 12:00 pm	140
	Generic (for all)	10Within 1 month, your cough and sputum production will decrease and you will be able to breathe easily. If you don't quit, your recovery will be delayed! Time: 2:00 pm	151
	Generic	11Within 6 months of quitting, you will be cured of TB as long as you take	139

[3] This is number of days to the quit date
[4] Tx is used for TB treatment and +1 means the first day of treatment

	(for all)	your medication. Don’t forget to take your TB medication regularly! Time: 5:00 pm	
	(smoking forms only)	12If you don’t quit you will have a higher chance of getting TB again! Craving to smoke is temporary but damage to your lungs is permanent! Time: 7:00 pm	137
	[5](chewing tobacco)	13If you don’t quit you will have a higher chance of getting TB again! Tobacco contains harmful substances and chewing can lead to mouth cancer! Time: 8:00 pm	147
Q (-5) Tx (+4)	Generic (for all)	14Tell your family and friends that you have TB so they can offer you support and help you to get better. TB does not spread by shaking hands, cuddling or hugging. Time: 9:00 am	161
	Generic (for all)	15Remember that your QUIT Date is near! 5 days to go! Share this date with your family/friends for support to stay tobacco free and continue taking TB medicine. Time: 12:00 pm	160
	Generic (for all)	16It is important to get your family and friends support during treatment. TB is spread through the air, so you will not transmit TB by sharing food or utensils. Time: 02:00 pm	160
	Generic (for all)	17Remember, there is **NO** safe or mild tobacco. Bidi, cigarette, hookah and chewing tobacco are all harmful. (Local tobacco product) is also harmful. Time: 05:00 pm	145
Q (-4) Tx (+5)	Generic (for all)	18It takes 6 months of regular treatment to kill TB germs but after few weeks of taking medicine you can continue life as normal. Take your TB !medication regularly Time: 9:00 am	163
	Generic (for all)	19.12 months after quitting your body will be highly resistant to catch TB again! Time: 12:00 pm	78
	Generic (for all)	20PREPARE TO GET RID OF TOBACCO PRODUCTS; matches, ashtrays, paan dan/chewing box, water-pipe, anything that might remind you of tobacco use. Time: 02:00 pm	141
	Generic (for all)	21Make a list of your reasons for quitting tobacco. Health, family, and saving money are good reasons to quit. Time: 05:00 pm	111
	(smoking forms only)	22Did you know that smoking can cause sexual problems? It can even cause impotence. Time: 07:00 pm	81
	(chewing tobacco)	23Did you know that chewing can cause staining of the teeth, bad breath and gum problems? It can even cause throat cancer. Time: 08:00 pm	120
Q (-3) Tx (+6)	Generic (for all)	24While preparing for your quit date, do not forget your TB medicines! Do not miss even a single dose, it will take longer to cure. Time: 9:00 am	129
	Generic (for all)	25When you first start using tobacco your brain changes and expects regular doses of nicotine. When you quit, nicotine is not available and	162

[5] Pink background is for chewing tobacco text messages

		you feel an urge for it. Time: 12:00 pm	
	Generic (for all)	26The urge for nicotine can be very strong and can undermine your motivation to quit, especially on occasions and situations that remind you of tobacco. Time: 02:00 pm	150
	Generic (for all)	27You are moving towards your QUIT date! 3 days to go! It is important that you request family and friends who use tobacco about not using it when they are with you. Time: 05:00 pm	163
	Generic (for all)	28Before your QUIT date, please think through things that tempt your desire to use tobacco, like certain times, or places or people. Write these situations down. Time: 07:00 pm	160
Q (-2) Tx (+7)	Generic (for all)	29Tobacco use will make it harder to cure your TB. Quitting will help you recover faster from TB as long as you take your medication. Time: 9:00 am	131
	Generic (for all)	30Did you think about situations that might trigger your desire to use tobacco? Start planning for how you will manage your urges to use tobacco. Time: 12:00 pm	144
	Generic (for all)	31To avoid the urges to use tobacco, think of alternative things you could do instead, like using gum, clove or cardamom to keep your mouth busy. Time: 02:00 pm	143
	Generic (for all)	32Many people use tobacco to deal with stress, tension or boredom. Try walking, doing exercise, yoga, deep breathing, meditation or prayer instead. Time: 05:00 pm	145
	Generic (for all)	33It is normal to be nervous about quitting. Don't panic, your family and friends are there to support you! Tell them that you are ready to quit for good! Time: 07:00 pm	152
Q (-1) Tx (+8)	Generic (for all)	34Remember, TB is curable if you take the advised treatment. However, outcome is poorer in tobacco users. Quitting will help you successfully recover from TB. Time: 09:00 am	157
	Generic (for all)	35Within first few hours of stopping tobacco use, your brain will start getting used to being without nicotine. This adjustment may result in craving or withdrawal. Time: 12:00 pm	162
	Generic (for all)	36Cravings can be uncomfortable and you might feel uneasy. Rest assured that these are perfectly normal and will reduce overtime as you progress with your quit. Time: 02:00 pm	158
	Generic (for all)	37After quitting you may feel constipated, irritable, restless or lightheaded due to withdrawal from nicotine that you may or may not get. These will go away soon! Consult with your TB Provider if needed. Time: 05:00 pm	161
	Generic (for all)	38Ready for quitting tomorrow! Hope you have removed all tobacco products, matches, ashtrays, paan dans, and waterpipe around you. If not do so right away! Time: 07:00 pm	153
QUIT	Generic	39You need clean air to fight against TB germs. Ask your family and	133

DATE (Q) This is the 8th day since joining mTB-Tobacco	(for all)	friends NOT to smoke around you. Continue taking your TB medicines! Time: 09:00 am	
	Generic (for all)	40Today is your QUIT DATE! Remove all tobacco products around you. You can do it! Time: 12:00 pm	78
	Generic (for all)	41Drink lot of fluids (water, coconut water, lime juice, lassi, buttermilk) today to feel fresh and fight cravings. Avoid sugary drinks or alcohol. Time: 02:00 pm	143
	Generic (for all)	42Have gum/cloves/cardamom/mints, water, and healthy snacks to get you through the day. You have made the right choice! Time: 05:00 pm	118
	Generic (for all)	43Tell your near and dear ones that you are trying to quit tobacco today. Ask for their support to stay tobacco free. Time: 07:00 pm	115
	Generic (for all)	44If you're feeling stressed or upset before sleeping at night, do not smoke tobacco. Tobacco use never solved any problem for you. You did it. THINK POSITIVE. Time: 09:00 pm	
Q (+1) Tx (+10)	Generic (for all)	45.24 hours tobacco free …well done! Be sure to reward yourself. Celebrate with your family and friends. Time: 09:00 am	101
	Generic (for all)	46The first days are going to be tough. Be patient. Don't switch from one kind of tobacco product to another. Do not use (local tobacco product). Time: 05:00 pm	143
	Generic (for all)	47Remind yourself of your reasons for quitting. That will help keep you on track when you need a boost. Time: 07:00 pm	101
Q (+2) Tx (+11)	Generic (for all)	48To be cured from TB you must remember to keep taking your medicines but having a healthy lifestyle will also help you recover better. Time: 09:00 am	135
	Generic (for all)	49It is important to eat healthy snacks (e.g., peanuts, raisins, aniseed) with proteins and vitamins frequently and in ample quantities. Make sure you get plenty of rest! Time: 12:00 pm	135
	Generic (for all)	50Refrain from consuming alcohol. Breathe clean fresh air and keep it tobacco free. Ask family and friends not to smoke indoors! Time: 02:00 pm	126
	Generic (for all)	51Refrain from using any kind of tobacco. Bidi, cigarette, hookah and chewing tobacco are all harmful. (Local tobacco product) is also harmful. Time: 05:00 pm	141
	Generic (for all)	52If your urge to smoke is because you're feeling stressed, try closing your eyes & taking deep breaths. Tell yourself that smoking is a false way of coping! Time: 07:00 pm	155
Q (+3) Tx (+12)	Generic (for all)	533rd day without tobacco! Your food will taste better. Drink plenty of water to avoid constipation. You may begin to feel better. Keep taking your TB medicine! Time: 12:00 pm	158
	Generic	54Talk to someone who has already QUIT or spend time with persons	87

	(for all)	who do not use tobacco. Time: 02:00 pm	
	Generic (for all)	55Keep your mouth busy! Use chewing gum, cloves, cardamom, or mints to avoid tobacco use. Time: 05:00 pm	87
Q (+4) Tx (+13)	Generic (for all)	564 days tobacco free! The first few days can be tough. Try to relax. Listen to music, have fresh fruits, do yoga or deep breathing. Time: 12:00 pm	130
	Generic (for all)	57Soon after starting your TB medicines, you may begin to feel better but you are not completely cured. You must keep taking your medicine! Time: 02:00 pm	138
	Generic (for all)	58Sometimes the TB medicine might have side-effects that make you want to stop taking it. Be patient, symptoms will subside. Just keep taking the medicines. Time: 05:00 pm	154
Q (+5) Tx (+14)	Generic (for all)	59Well done for staying tobacco free and regularly taking your TB medicine. Your body is already starting to heal. Keep up the good work. Time: 09:00 am	136
	Generic (for all)	60Day 5! Stay positive. Do not let things get you down. The journey towards a tobacco and TB free life is definitely worth it. Time: 12:00 pm	124
	Generic (for all)	61Drink lots of fluids like lime juice, buttermilk, coconut water. Avoid cold drinks and alcohol. Time: 02:00 pm	95
	Generic (for all)	62Stay away from people and places that remind you of tobacco. Do things which help you to relax. Some people find gardening, sewing, or watching a movie helpful. Time: 05:00 pm	160
Q (+6) Tx (+15)	Generic (for all)	63Get your family and friends support as your body combats TB and you deal with cravings. Ask a family member or friend to remind you to take your medicines. Time: 09:00 am	155
	Generic (for all)	64During festivals it is easy to forget taking medicines. Ask your friends and family members to remind you. Always carry your medicines while traveling! Time: 12:00 pm	153
	Generic (for all)	65. 6 days tobacco free! Your body is free from tobacco related toxins. Stay active or go for a walk; it helps to reduce stress. Time: 02:00 pm	126
	Generic (for all)	66If you’re feeling stressed or upset, talk to someone close to you for support. Time: 05:00 pm	78
Q (+7) Tx (+16)	Generic (for all)	67, 1 week tobacco free! Do not look back now. Feel happy and do something special today to celebrate your success! Time: 09:00 am	111
	Generic (for all)	68Find out how much money you have saved from not using tobacco in this week. Think about what else you can do with the saved money! Time: 12:00 pm	131
	Generic (for all)	69If you have the urge to use tobacco, watch a movie, go out or buy yourself something as reward for staying QUIT. Time: 05:00 pm	112
Q (+8) Tx (+17)	Generic (for all)	70Remember to keep coming to see the TB provider for your appointments or if you are having any problems. You will recover from	162

		TB if you keep taking your medicine. Time: 09:00 am	
	Generic (for all)	71Use a calendar to check off the days you have taken your TB medicines and the days that you have stayed tobacco free! Time: 05:00 pm	117
Q (+9) Tx (+18)	Generic (for all)	72Quitting tobacco makes you feel healthier and energetic. Time: 09:00 am	55
	Generic (for all)	73Tobacco smoke is dangerous for everyone, including non-smokers who inhale this smoke. Avoid places where others are smoking. Time: 12:00 pm	124
	Generic (for all)	74Smoking spreads TB germs. Make your home tobacco free! Time: 05:00 pm	54
Q (+10) Tx (+19)	Generic (for all)	75Hope you are feeling good! Keep up that positive attitude and stay strong. Stay away from all tobacco products, even those prepared locally. Time: 09:00 am	141
	Generic (for all)	76Tobacco use does not make anything better. Stay strong and say NO if anyone asks you to smoke/chew tobacco. Be proud that you said NO. Time: 05:00 pm	134
Q (+11) Tx (+20)	Generic (for all)	77Eat nutritious and healthy food. It will help you stay well and get better faster. Remember to keep taking your TB medicine! Time: 09:00 am	124
	Generic (for all)	78Avoid fried food when you have craving. Eat vegetable, salad, fruit, drink lassi, milk, coconut water. Time: 05:00 pm	150
Q (+12) Tx (+21)	Generic (for all)	79Exercise regularly, it helps to strengthen your body defences against TB germs. Time: 09:00 am	79
	Generic (for all)	80Gaining a little weight after quitting may occur for some time, but healthy eating and exercise can prevent this. Time: 05:00 pm	113
Q (+13) Tx (+22)	Generic (for all)	81Remember to get more medicines you run out! Check your next appointment date with TB provider and mark the calendar! Time: 09:00 am	123
	Generic (for all)	82Stay positive by exercising, doing yoga, meditation or starting a new hobby. Make friends with people who do not use tobacco! Time: 02:00 pm	125
	Generic (for all)	83Adopt a healthy lifestyle. Reduce your chances of ever developing TB again. Time: 05:00 pm	75
Q (+14) Tx (+23)	Generic (for all)	84Well done, 2 weeks tobacco free! How much time, energy, health, and money you have saved! You deserve a treat-make today special! Time: 09:00 am	129
	Generic (for all)	85Nobody likes a dirty mouth and bad breath. Now your smile is brighter and your mouth healthier. Time: 02:00 pm	95
	Generic (for all)	86Remind people close to you that you have QUIT tobacco. They want only what is best for you and that means staying QUIT and healthy. Time: 05:00 pm	131
Q (+15) Tx (+24)	Generic (for all)	87Congratulations! Being tobacco free means you are not a slave to something that actually harms you. Time: 09:00 am	99
Q (+16)	Generic	88Well done for taking your TB medicines regularly. You must be feeling	141

Tx (+25)	(for all)	much better now. Your next monthly follow-up appointment is in 5 days! Time: 09:00 am	
	Generic (for all)	89, 16 days! Tobacco use never solved any problem for you. YOU DID IT. You can do great things, so think positively. Time: 12:00 pm	111
Q (+17) Tx (+26)	Generic (for all)	90Share your TB experience with others and motivate them to seek help if needed! Ask your near ones to quit tobacco use as it increases their risk of getting TB. Time: 09:00 am	159
	Generic (for all)	91Keep up the great effort! Spend time with someone close to you. Go to places where no one smokes or uses tobacco. Time: 02:00 pm	113
	Generic (for all)	92Avoid people or places that remind you of using tobacco or going near shops that sell tobacco. It will make coping easier. Time: 05:00 pm	122
Q (+18) Tx (+27)	Generic (for all)	93Your next monthly follow-up appointment is in 3 days! Do not forget to visit your TB provider and collect more medicine. Time: 09:00 am	120
	Generic (for all)	94If you are bored, find something to do, so you won't feel tempted to use tobacco. Plan an outing, spend time with friends and family. Time: 02:00 pm	133
	Generic (for all)	95You used to pay money for something that would kill you. Be happy you are not a target anymore. Time: 05:00 pm	95
Q (+19) Tx (+28)	Generic (for all)	96Encourage your friends and family to get tested for TB, if they live with a TB patient. Remember, TB spreads through the air. Time: 09:00 am	125
	Generic (for all)	97Remember, change is POSSIBLE and for your good. Do not give up your resolve to QUIT tobacco. Even if you fall, get up and try again. Time: 05:00 pm	132
Q (+20) Tx (+29)	Generic (for all)	98Lung capacity increases by 30% after a few weeks without smoking! Breathe deeply and take a walk. Time: 09:00 am	96
	Generic (for all)	99Distract yourself during a craving by being active. Physical activity can help you stay QUIT and improves your mood. Time: 05:00 pm	116
Q (+21) Tx (+30)	Generic (for all)	100. 3 weeks tobacco free! Reward yourself again. Go for a walk with someone close, buy a small gift or cook something nice. Time: 09:00 am	119
	Generic (for all)	101And well done for successfully completing 1 month of TB treatment! Your monthly follow-up appointment is today, please do not forget to go! Time: 09:30 am	139
Q (+22) Tx (+31)	Generic (for all)	102Be a role model; motivate your friends and family members to quit tobacco use and get tested for TB if they live with a TB patient. Time: 09:00 am	131
	Generic (for all)	103Don't be fooled by tobacco and (local tobacco product) advertisements and promotion. You now know the REAL truth about smoking & other tobacco use. Tell others. Time: 05:00 pm	160
Q (+23) Tx (+32)	Generic (for all)	104Hope you collected more medicine for your TB treatment. Write yourself a note and put it on the fridge or a door. Keep taking medicines and stay healthy!	153

		Time: 12:00 pm	
	Generic (for all)	105The majority of tobacco users wish they had never started. Be happy with your decision to QUIT tobacco. Time: 02:00 pm	103
	Generic (for all)	106All forms of tobacco are harmful. Remember, tobacco use causes high blood pressure, heart attack, TB, paralysis and cancers. Time: 05:00 pm	124
Q (+24) Tx (+33)	Generic (for all)	107Continue taking your TB medicines! In case, you missed your dose, take the next dose as scheduled. Time: 09:00 am	97
	Generic (for all)	108Quitting tobacco not only helps your lungs heal, save you from cancer but makes you healthier and stronger. Time: 05:00 pm	108
Q (+25) Tx (+34)	Generic (for all)	109Ask a person close to you how your breath is after Quitting. Your breath is fresh! No bad breath! Aren't you glad you QUIT? Time: 02:00 pm	123
Q (+26) Tx (+35)	Generic (for all)	110Be careful when you go out. Do not let yourself slip and use tobacco again. You have come so far. Remember to take your TB medicine! Time: 02:00 pm	131
Q (+27) Tx (+36)	Generic (for all)	111Don't forget to reward yourself for staying tobacco free! Watch a movie, buy something, or put away the money you have saved in a money box. Time: 02:00 pm	140
Q (+28) Tx (+37)	Generic (for all)	112Alcohol may increase your urge to use tobacco and cause a relapse. Stay away from alcohol, cold drinks and (local tobacco products). Time: 02:00 pm	132
	Generic (for all)	113Alcohol also damages your liver and your body's defence against infections such as TB and HIV. Cutting down on alcohol will help recover faster from TB. Time: 05:00 pm	152
Q (+29) Tx (+38)	Generic (for all)	114Don't let stress undo all your hard work. You CAN deal with it without tobacco. Go for walk, exercise, take deep breaths. Stay healthy, keep taking TB medicine! Time: 02:00 pm	159
Q (+30) Tx (+39)	Generic (for all)	115Congrats! 1 MONTH since you became tobacco free! It is a major milestone. Share the good news with your family and friends. Do something special! Time: 12:00 pm	145
Q (+31) Tx (+40)	Generic (for all)	116Tobacco users face an increased risk of sexual problems. Be glad you have decided to QUIT. Stay tobacco free! Time:12:00 pm	109
Q (+32) Tx (+41)	Generic (for all)	117If you stop taking TB medicines, you can pass TB germs to others! Keep taking your medicine to help you recover and protect others from TB! Time: 12:00 pm	139
Q (+33) Tx (+42)	Generic (for all)	118BREATHE DEEPLY! Inhale through nose, exhale through mouth, repeat 10 times. This will help you get over the urge for tobacco. Time: 12:00 pm	125
Q (+34) Tx (+43)	Generic (for all)	119All kinds of tobacco reduce life span and make you look aged. Smoking or chewing tobacco is NOT safe. Time: 12:00 pm	101
Q (+35)	Generic	120By taking TB medicines you can prevent TB disease and keep your	116

Tx (+44)	(for all)	family healthy! Don’t forget to take your medicines! Time: 09:00 am	
	Generic (for all)	121Even now, some things may make you want to use tobacco again! Use the skills you have learned to get through your craving without tobacco. Time: 05:00 pm	138
Q (+36) Tx (+45)	Generic (for all)	122When faced with any difficulty, try to overcome it. Quitting tobacco is no different. Do not look back now. You have come a long way. Time: 02:00 pm	133
Q (+37) Tx (+46)	Generic (for all)	123Congrats! It’s been 1.5 months since you started TB treatment. Continue your commitment with taking TB medicines regularly! Time: 09:00 am	123
	Generic (for all)	124Think of what tobacco use costs you in one year. Think about all the other things you could have done with that money. Keep up the tobacco free lifestyle! Time: 12:00 pm	154
Q (+38) Tx (+47)	Generic (for all)	125Arguments may still trigger desire for tobacco. To get relief from stress, practice deep breathing and meditation. Time: 12:00 pm	114
Q (+39) Tx (+48)	Generic (for all)	126If you slip and use tobacco again, keep trying to QUIT. Learn from your mistakes this time. Time: 12:00 pm	91
Q (+40) Tx (+49)	Generic (for all)	127Remember, if you stop taking your TB medicines or don’t take it correctly, your medicine may lose the power to cure your TB! Time: 09:00 am	125
	Generic (for all)	128Pregnant women who use tobacco or are exposed to others’ tobacco smoke are at risk for stillbirths & low weight babies. Time: 05:00 pm	79
Q (+41) Tx (+50)	Generic (for all)	129Remember to take your TB medicine in full dose, even when you don’t feel good. It is the only way to kill TB germs. Time: 09:00 am	115
	Generic (for all)	130This is the beginning of a new life for you. Now you know the way to succeed. Stay tobacco free and help others in quitting tobacco! Time: 02:00 pm	132
	Generic (for all)	131Now that you are tobacco free, encourage others to QUIT- Time: 05:00 pm	56
mTB-Tobacco Follow-up phase!			
Q (+52) Tx (+61)	Generic (for all)	132Well done! You have successfully completed first 2 months of your TB treatment. Now it’s time to visit your provider for sputum test and collect more medicine! Time: 12:00 pm	159
Q (+112) Tx (+121)	Generic (for all)	133Well done! You have successfully completed 4 months of your TB treatment. Now it’s time to visit your provider for sputum test and collect more medicine! شاباش! آپ نے ٹی بی کے علاج کے 4 مہینے کامیابی سے مکمل کر لیے ہیں۔ اب ڈاکٹر کے پاس جا کر تھوک کا ٹیسٹ کروانے اور مزید دوائیں لینے کا وقت ہے۔ Time: 12:00 pm	153
Q (+180) Tx (+189)	Generic (for all)	134Well done! You have successfully completed 6 months of your TB treatment. Now it’s time to visit your provider for the final sputum test. Time: 12:00 pm	137

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